A Latent Oscillator Measurement Model to Simulate Emotional-Expression Score Dynamics in Video

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Abstract

Automated vision models turn video into dense time series of emotion scores. Real score series alone cannot show whether a method for estimating dimensions would find the right structure if that structure were actually present. We introduce the Latent Oscillator Measurement Model (LOMM), a simulation framework with three layers that can be varied independently: latent dynamics, time-varying expression, and a measurement model that turns latent states into observed scores. LOMM generates either bounded scores that sum to one, like classifier probabilities, or continuous indicators with a known factor structure. Study 1 compared simulated with real score series from 100 MAFW video clips using five-fold cross-validation. Letting video clips differ in baseline score level and overall scale reduced the mean discrepancy from 0.3250 to 0.2567, a 21.0% improvement. Study 2 tested how well four methods recover the number of dimensions from continuous data with a known factor structure. With Louvain community detection, 100 observations per clip, and failed or timed-out fits counted as incorrect, accuracy was 0.939 for DynEGA, 0.884 for static EGA, 0.777 for GraphicalVAR, 0.176 for GIMME, and 1.000 for an oracle given the true structure. A population analysis shows that networks built from derivatives and networks built from lagged residuals share the same $1/(k-1)$ upper bound on within-dimension partial correlations, so using derivatives carries no automatic advantage. Shorter time series and independent starting states reduced accuracy, whereas gradual rather than abrupt changes in expression mattered little. The two studies answer different questions, and Study 2 does not establish recovery from bounded scores. LOMM lets researchers test dynamic methods against a known structure before treating patterns in classifier scores as substantive findings.

Introduction

Video is increasingly used as a source of intensive behavioral data because it preserves the temporal ordering of visible action and can be analyzed at a resolution that manual frame-by-frame coding cannot sustain. Deep-learning systems can now convert frames or short video segments into dense time series of model-derived emotional-expression scores (S. Li & Deng, 2022; Meng et al., 2017). In a facial-expression recognition task, a model maps each image or segment to category probabilities or continuous scores for labels such as happiness, anger, sadness, fear, surprise, disgust, or neutral expression.

These score streams are valuable because many behavioral questions concern how visible affect unfolds over time. Researchers may ask whether expressive states persist or switch, whether they return after perturbation, and whether many model-output channels reflect a smaller number of expressive dimensions. Such questions arise in psychotherapy, autism assessment, collaborative work, public communication, and social-media video (Halamová et al., 2023; Holiday et al., 2023; Q. Li et al., 2024; Manelis-Baram et al., 2025).

Automated facial-analysis systems now make this kind of work feasible at scale. Datasets such as FER2013 supported deep-learning approaches to facial-expression recognition, especially convolutional neural networks (CNNs) (Arriaga et al., 2017; Ezerceli & Eskil, 2022; Goodfellow et al., 2013). Most early systems inherited emotion categories from their training data and returned scores for fixed labels such as anger, disgust, fear, happiness, sadness, surprise, and neutral facial expression (Wang et al., 2022). CLIP-based and related vision-language models also support zero-shot classification with user-specified labels (Foteinopoulou & Patras, 2023; H. Li et al., 2023; Radford et al., 2021; Tomašević et al., 2026). Each frame can therefore be represented by a vector of prompt-defined classifier scores.

Facial-expression benchmarks also show why these scores require task-specific validation. FER-adapted vision-language models such as CLIPER and EmoCLIP improve on an unadapted CLIP baseline by adding expression-specific text descriptions or training objectives (Foteinopoulou & Patras, 2023; H. Li et al., 2023). EmoCLIP, for example, reported gains exceeding 10 percentage points in weighted average recall and 5 points in unweighted average recall over CLIP on several zero-shot video benchmarks. Such gains indicate that performance depends on model adaptation, the label set, prompt construction, and the evaluation dataset. These accuracy studies do not establish calibration of arbitrary prompt-defined scores, and the eleven-channel scoring procedure examined here has no frame-level label-validity claim.

The central obstacle is measurement. Emotional-expression scores are trained-model outputs that index visible behavior through a scoring procedure. Facial-expression technologies may be used for emotion recognition or for social-interaction analysis, and these interpretations answer different questions (Allison & McKeown,

2023). A further psychometric question is whether multiple score channels should be analyzed separately or treated as fallible indicators of lower-dimensional expressive processes (Mehu & Scherer, 2015; Nesselroade et al., 2002; Russell, 2003). Measurement error can alter the structure recovered by dynamic methods (Kankaanpää et al., 2026). Method evaluation must therefore separate latent temporal organization from the process that turns it into observed scores.

This requirement raises a practical question that empirical data cannot answer: would a recovery method find the intended structure if it were present? Answering it requires a generator in which both latent dynamics and the multichannel observation process are known before estimation. Oscillator models provide parameters for level, velocity, feedback, and perturbation in intraindividual dynamics (Boker, 2015; Boker & Nesselroade, 2002; Ollero et al., 2025). A useful validation design must also vary dimensionality, indicator blocks, measurement quality, and activity while preserving a known recovery target.

This paper introduces the Latent Oscillator Measurement Model (LOMM), a simulation tool that generates emotional-expression score series whose structure is known in advance. LOMM separates three operations. A latent layer generates the underlying trajectories, an activity layer controls which dimensions are expressed at each moment, and an observation layer turns those trajectories into noisy scores on many channels. Researchers can specify the number of latent dimensions, the dynamic parameters, loading strength, measurement error, the activity pattern, how much video clips differ from one another, and whether the simulated scores are bounded or continuous. All of these are fixed before any method is applied, so what a method recovers can be compared with what generated the data.

LOMM integrates established modeling ideas in a single generative design. Differential-equation models represent level and velocity, continuous-time state-space models motivate the exact transition between observation times, and dynamic factor models motivate the multivariate measurement structure (Boker, 2015; Boker & Nesselroade, 2002; Nesselroade et al., 2002; Zhang et al., 2008). The contribution lies in their measurement-first integration: latent dynamics, activity, and observation geometry can be varied independently while the indicator partition remains known.

We use this framework in two linked studies. Study 1 asks whether simulated score series reproduce the statistical properties of real classifier scores from 100 MAFW video clips, evaluated with five-fold cross-validation. Study 2 asks whether four methods recover the correct number of dimensions, and the correct assignment of variables to dimensions, from simulated continuous data with a known factor structure. A follow-up sensitivity analysis tests whether recovery changes when video clips differ in their baseline score levels and overall variability. Throughout the paper we keep two things separate: whether simulated data

look like real data, and whether a method recovers a structure we planted ourselves.

A population-level analysis compares the networks these methods estimate on a common scale. When all loadings are equal, the derivative-based network estimated by DynEGA and the lagged-residual network estimated by GraphicalVAR approach the same upper bound on within-dimension partial correlations, $1/(k-1)$ for k indicators per dimension. Which method does better therefore depends on the signal-to-noise ratio and the length of the time series, not on whether a network is built from derivatives or from lagged residuals.

Present Research

Dynamic Exploratory Graph Analysis [DynEGA; H. Golino et al. (2022)] is the focal method because it was designed to find groups of variables whose changes are coordinated over time. GraphicalVAR and GIMME are neighboring methods for intensive longitudinal data; each estimates a different structure and turns it into a network in a different way. Static EGA, which ignores the time ordering entirely, shows how much can be recovered from score levels alone. Finally, an oracle receives the true structure and passes it through the same community-detection step as the other methods, which confirms that the shared machinery works as intended.

The paper addresses three questions. First, how closely do simulated score series reproduce the statistical properties of real classifier scores, judged on video clips held out from calibration? Second, how well do DynEGA, static EGA, GraphicalVAR, and GIMME recover a known factor structure from simulated continuous data? Third, how does that recovery change under shorter time series, independent starting states, gradual rather than abrupt changes in expression, cross-loadings, correlated residuals within a dimension, correlated inputs across dimensions, and bounded scores? The oracle answers none of these questions on its own; it is an implementation check rather than a target that a real estimator could be expected to reach.

The two studies answer different questions and support different conclusions. Study 1 asks only whether simulated data resemble real data on a set of chosen summary statistics; it says nothing about whether any structure can be recovered. Study 2 asks only whether methods recover a structure we planted; it says nothing about whether the simulated data look like real classifier output. The sensitivity analysis sits inside Study 2: it adds between-clip differences in baseline and scale while keeping the data-generating factor structure fixed. Figure 1 shows how the two studies relate.

Appendix B gives the population-level argument behind Study 2. Under ideal measurement, indicators belonging to the same dimension share a common signal, and indicators belonging to different dimensions are uncorrelated in the population. Weaker loadings, larger measurement error, less expressive activity, and

dependence across dimensions all shrink the separation a method has to work with. The simulations then test departures from that ideal, and they always distinguish whether a method produced a usable result at all from how accurate it was when it did.

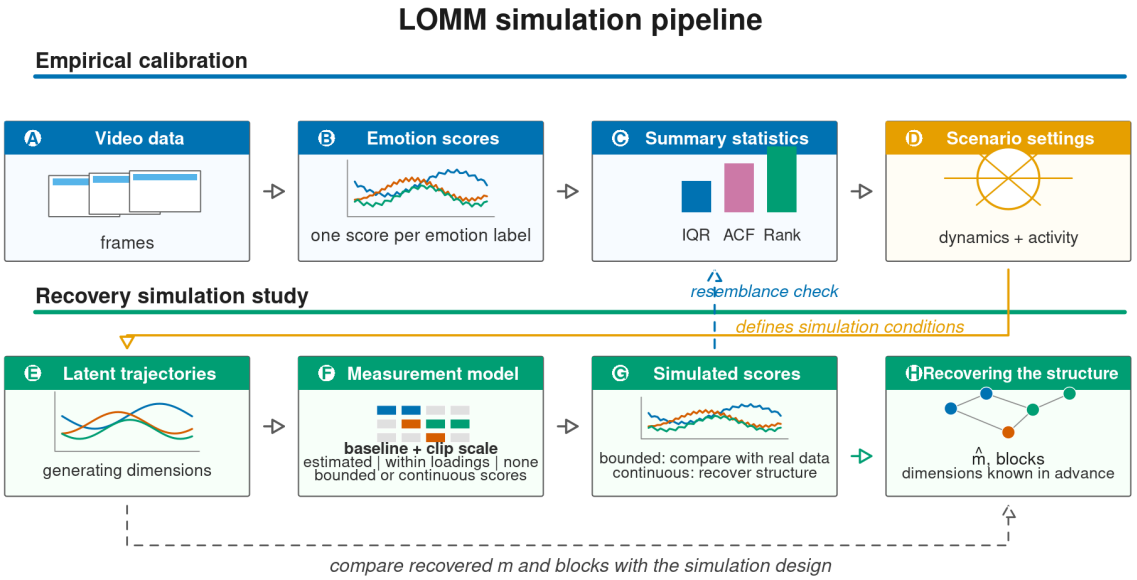


Figure 1: Overview of the two LOMM studies. Panels A–C show Study 1, in which simulated score series are compared with real MAFW classifier scores on video clips held out from calibration. Panel D shows the latent dynamics and expression patterns shared by both studies. Panels E–G show that Study 1 uses bounded scores that sum to one, whereas Study 2 uses continuous indicators. Panel H shows Study 2, in which each method's recovered dimensions and variable assignments are compared with the structure that generated the data. Study 2 does not test recovery from bounded scores.

The Latent Oscillator Measurement Model

LOMM starts from a measurement problem. A video yields one score per emotion label per frame, and the simulation treats those scores as fallible indicators of a smaller number of underlying expressive processes. Over a short clip, expression may stay low, rise briefly and subside, persist, or alternate between states. A useful latent representation therefore has to describe a current level, a rate of change, a pull back toward rest, overshoot, damping, and random perturbation, all of which are central to research on affective dynamics (Hollenstein, 2007; Houben et al., 2015; Kuppens & Verduyn, 2017).

Oscillator models describe exactly these features. Prior work has used them for intraindividual dynamics in which a current level and its rate of change jointly determine what happens next (Boker, 2015; Boker & Nesselroade, 2002; Ollero et al., 2025). LOMM builds on that representation and adds two further layers, giving three in total: a latent layer that generates expressive trajectories, an activity layer that decides which dimension is expressed at each moment, and an observation layer that turns those trajectories into observed

scores, either bounded scores that sum to one or continuous indicators with a known factor structure (H. Golino et al., 2022; Nesselroade et al., 2002). The three layers can be varied independently, which is what allows a simulated dataset to be attributed to a specific part of the generating process. This section describes each layer with the minimum notation needed to follow the studies; Appendix A gives the complete formal specification.

The latent layer treats each expressive dimension as a damped oscillator observed at equally spaced times. Let $n = 1, \dots, T$ index observations, let $\Delta > 0$ be the interval between them, and define physical time as $t_n = (n-1)\Delta$. Video i and dimension f have a state $\mathbf{s}_{if,n} = [x_{if,n}, v_{if,n}]^\top$ holding the current level and its rate of change. The state moves forward one step according to

$$\mathbf{s}_{if,n+1} = \Phi_f(\Delta)\mathbf{s}_{if,n} + \Gamma_f(\Delta)q_{if,n}, \qquad q_{if,n} \sim \mathcal{N}(0, \sigma_{qf}^2). \tag{1}$$

Two parameters govern each dimension, and they are collected in the feedback matrix

$$\mathbf{A}_f = \begin{bmatrix} 0 & 1 \\ \eta_f & \zeta_f \end{bmatrix}.$$

Position feedback η_f sets how strongly the process is pulled back toward its resting level, and velocity feedback ζ_f sets how quickly movement is damped. Together they determine the natural period of an oscillation and how fast an expressive episode fades. The matrix $\Phi_f(\Delta) = \exp(\mathbf{A}_f\Delta)$ carries the state forward deterministically across one observation interval, and the vector $\Gamma_f(\Delta)$ distributes the random acceleration input $q_{if,n}$ across the level and its rate of change. Because these two quantities are computed exactly rather than approximated by a discretization, simulated trajectories are correct at any observation interval Δ. Appendix A gives their closed forms.

The acceleration inputs are what keep a trajectory from settling into a smooth decay: they perturb the process at every step, producing the continuing peaks, reversals, and irregular movement visible in real expression. Their standard deviation is σ_{qf}. They are independent across videos, dimensions, and time steps, and they are independent of initial states, activity schedules, and measurement error.

The sign of the two feedback parameters determines the regime. When both are negative, the dimension is stable: it oscillates and settles back toward its resting level. When velocity feedback is positive, movement is amplified instead, and the trajectory grows over the observed record. We include such conditions deliberately as stress tests rather than as plausible affective regimes, and we report a diagnostic for how large the resulting

trajectories become (Appendix A gives the stability conditions and the amplification diagnostics; Table 2 lists which scenarios use which regime).

Dimensions are uncoupled: η_f, ζ_f, and σ_{qf} may differ across dimensions, but no dimension drives another. Study 2's main conditions start every dimension of every video from the same state ($x_{if,1} = 0.25$, $v_{if,1} = 2$), and a sensitivity analysis instead draws starting states independently for each video and dimension. This distinction matters more than it might appear: a shared starting state creates a transient that every dimension has in common, which is itself recoverable structure. Appendix A gives the stationary distribution used for the independent starts.

The activity layer decides which dimension is actually being expressed at each moment. It multiplies each latent trajectory by a nonnegative activity value, giving $\tilde{x}_{if,n} = a_{if,n} x_{if,n}$, and this product is what the observation layer sees. The activity values follow a scenario-specific schedule built from an active level, a low background level, and dwell times that determine how long each state lasts. In single-dominance scenarios, one dimension stays highly expressed for the whole clip. In switching scenarios, the expressed dimension changes after sampled dwell intervals. In low-expression scenarios, every dimension stays near the background level.

Whether activity changes abruptly or gradually is worth separating from the latent dynamics, because an abrupt switch is itself a large change in the observed series. A method that works with derivatives will see a spike at that switch even though the underlying oscillator did not accelerate. We therefore run a sensitivity analysis that replaces each abrupt switch with a five-observation gradual transition while holding the dimensions and their indicators fixed. Appendix A gives the exact identity separating the contribution of the changing activity from the contribution of the changing trajectory.

The observation layer turns the activity-scaled trajectories into observed scores. It does so in two steps: it first builds a vector of coordinates from the latent positions, and then places that vector on whichever scale the study requires. For video i at observation n, the first step is

$$\mathbf{u}_{in} = \mu + \mathbf{b}_i + \kappa s_i \left(\Lambda \tilde{\mathbf{x}}_{in} + \varepsilon_{in}\right), \qquad \varepsilon_{in} \sim \mathcal{N}(\mathbf{0}, \Sigma_i). \tag{2}$$

This is an ordinary factor model with two additions. The familiar parts are the loading matrix Λ, which maps the m activity-scaled latent positions onto the observed coordinates, the measurement error ε_{in}, and a fixed center μ shared by all videos. The measurement errors are independent over time and across videos, and independent of everything in the latent and activity layers. The additions are the two video-specific

terms that let clips differ from one another: the baseline $\mathbf{b}_i$ shifts a whole clip's scores up or down, and the scale $s_i > 0$ stretches or compresses both its signal and its noise. A global scale $\kappa > 0$ multiplies the same quantities for every clip.

These two terms are the switch between the studies. Study 2's main conditions turn them off, setting $\mathbf{b}_i = \mathbf{0}$ and $s_i = 1$ so that every video shares one measurement model and the factor structure is the only source of shared variance. Study 1 turns them on and estimates them from data: the baseline is drawn from a distribution whose covariance is estimated from training clips, and the scale is drawn as $s_i \sim \text{Unif}(0.5, 1.5)$, which averages one. The sensitivity analysis in Study 2 turns them on in a restricted way, drawing baselines that lie within the space spanned by the loadings so that between-clip differences stay consistent with the factor structure rather than adding a new source of variance. Appendix A gives all three constructions.

Study 1's calibrated settings are a natural period of 12 observations, a damping ratio of 0.5, $\kappa = 1$, and a measurement-error standard deviation of 0.25 in the contrast coordinates. Expressed as feedback parameters, these give $\eta_f = -0.2742$ and $\zeta_f = -0.5236$, with an acceleration-input standard deviation of $\sigma_{qf} = 0.5418$ after scaling the trajectories to unit variance. Study 1's selected configuration holds activity constant at one; a separate variant adds a brief three-observation transition.

The second step places the coordinates on the reported scale, and this is where the two studies diverge. Study 2 uses the coordinates directly as continuous indicators, so each of the $p = mk$ observed variables loads on exactly one dimension and carries its own measurement error. This is the standard continuous factor model that dimensionality methods are designed for, which is why it is the setting in which we ask whether they recover a known structure. Study 1 instead needs output that looks like a classifier's, so LOMM maps the coordinates onto J strictly positive scores that sum to one:

$$\mathbf{p}_{in} = \text{softmax}\left(\frac{\mathbf{V}\mathbf{u}_{in}}{\tau}\right). \tag{3}$$

The softmax function exponentiates its argument and divides by the total, which is how a classifier turns raw outputs into scores that sum to one. The fixed matrix $\mathbf{V}$ expands the $d = J - 1$ coordinates of $\mathbf{u}_{in}$ into J channels, and the temperature $\tau > 0$ controls how sharply the scores concentrate on a single channel. The construction has a property we rely on: because the columns of $\mathbf{V}$ are orthonormal and sum to zero, the log-ratio coordinates of $\mathbf{p}_{in}$ recover $\mathbf{u}_{in}/\tau$ exactly. The only thing lost in the map is a level common to all channels, which no set of scores constrained to sum to one could reveal in any case. Appendix A gives the log-ratio definitions, the construction of $\mathbf{V}$, and the full identification argument.

Study 2's continuous indicators are generated before this map is applied. Each of the $p = mk$ indicators has one primary loading on its own dimension and its own measurement error. So that indicators within a dimension are not identical, loadings are perturbed by $\text{Unif}(-0.15, 0.15)$ and residual variances by $\text{Unif}(-0.01, 0.01)$; because the smallest nominal loading in the design is 0.40 and the smallest nominal residual standard deviation is 0.125, these perturbations leave every loading and variance positive.

Whichever scale is used, the simulated channels represent classifier output about visible expression and nothing more. LOMM tests whether methods recover a measurement process we specified in advance under controlled loading strength, measurement error, dynamic regime, activity pattern, and between-clip variability. Whether the emotion labels attached to those channels are valid, and whether they correspond to subjective emotional experience, are separate questions the simulation cannot address. A neutral channel can appear in real classifier output, but it is never assigned to a latent dimension here unless a scenario deliberately specifies one.

To make this concrete, a three-dimensional LOMM could stand for approach-affiliation, threat-defense, and orienting-novelty. Score channels for happiness, amusement, trust, or smiling would then be indicators of approach-affiliation; channels for anger, fear, disgust, contempt, or anxiety would be indicators of threat-defense; and channels for surprise, anticipation, confusion, or attention would be indicators of orienting-novelty. These labels are an example of how a researcher might set up a simulation, not validated affect categories in the real MAFW scores. The example draws on affect taxonomies that distinguish broad dimensions, discrete expression families, and opponent tendencies (Plutchik, 2001; Russell, 1980, 2003; Russell & Mehrabian, 1977), and it is consistent with facial-expression benchmarks that combine categorical expression labels with dimensional ratings (Kollias et al., 2023; Mehu & Scherer, 2015; Mollahosseini et al., 2019). A different taxonomy can be encoded by changing which channels belong to which dimension, leaving everything else about LOMM unchanged.

What LOMM Can and Cannot Identify

Knowing the settings that generated a dataset is not the same as being able to recover them from data, and LOMM is a generator rather than a model to be fitted. Several of its quantities are not separately identifiable even in principle. Dimensions are recoverable only up to a relabeling. The size of a loading can always be traded against the scale of the latent trajectory it multiplies, so neither is pinned down without an explicit constraint, and activity gain and acceleration-input variance both feed latent amplitude in the same confounded way. The bounded map adds two more: shifting every channel's input by a common constant leaves the scores unchanged, and the temperature can be absorbed into the scale of the coordinates.

The sampling rate imposes one further limit. An oscillation faster than half the sampling rate is indistinguishable from a slower one, so the observation interval sets a ceiling on the frequencies that could ever be identified. Every oscillation used here falls below that ceiling with room to spare. Study 1 uses $\Delta = 1$, giving the most restrictive bound of approximately 3.14 against a largest angular frequency of approximately 0.78; Study 2 uses $\Delta \in \{0.25, 0.4\}$, giving bounds of approximately 12.57 and 7.85 against a largest angular frequency of approximately 1.04. Fitting LOMM to real data would require constraints on factor sign and scale, on the activity schedule, on the loading or input scale, on the temperature, and on the frequency range; Appendix A states each of these formally. The simulations reported here estimate none of these parameters. They ask only what methods recover from data whose structure we fixed in advance.

Figure 2 shows the same three-dimension example twice to make the role of the random acceleration inputs visible.

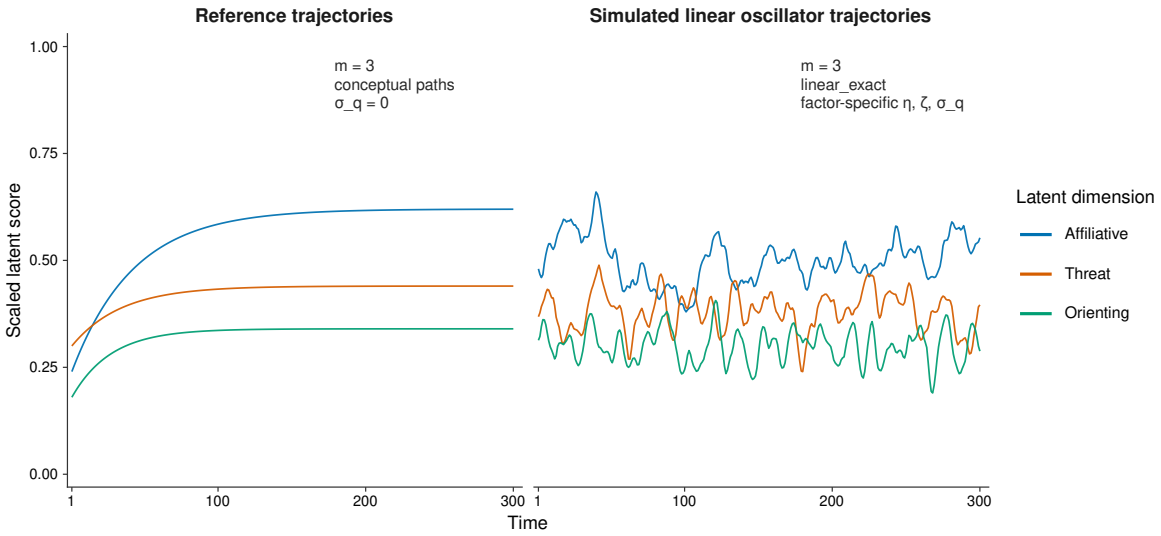


Figure 2: A three-dimensional LOMM example generated with Equation 1. On the left, the acceleration inputs are fixed at zero, so each dimension simply oscillates and decays. On the right, the same feedback parameters and starting states are used with random acceleration inputs.

Figure 2 isolates what the random acceleration inputs contribute. With the inputs fixed at zero, the three stable dimensions oscillate and decay toward a common resting level. Adding random inputs to the same transition produces the continuing peaks, reversals, and irregular movement that make real expression series look the way they do. The figure shows latent trajectories only; persistent differences in the level of observed scores enter later, through the center and the video-clip baseline in Equation 2.

Table 1: What each analysis simulates and what it supports. Study 1 asks whether simulated data resemble real data; Study 2 asks whether methods recover a structure specified in advance.

Analysis	Data type and configuration	Target	Methods	What it supports
Study 1: five-fold comparison	Bounded scores summing to one; clips differ in baseline and scale	Chosen summary statistics of real score series	Comparison of eight configurations	Similarity to real data in this sample
Study 2: main benchmark ($T = 100$)	Continuous indicators; all clips share one measurement model	Number of dimensions and assignment of variables to them	DynEGA, static EGA, GraphicalVAR, GIMME, oracle	Recovery under the data-generating factor structure
Study 2: shorter series ($T = 25$)	Continuous indicators; all clips share one measurement model	Number of dimensions and assignment of variables to them	DynEGA, static EGA, GraphicalVAR, GIMME, oracle	Recovery when each clip has fewer observations
Study 2: targeted departures	Continuous indicators; all clips share one measurement model	Number of dimensions and assignment of variables to them	DynEGA, static EGA, GraphicalVAR, oracle	Sensitivity to starting states, activity transitions, loadings, residuals, and inputs
Study 2: bounded-score departure	Bounded scores summing to one, or their log-ratio coordinates	Dimensions and variable assignment among the original channels	DynEGA, static EGA, GraphicalVAR	Recovery once scores are constrained to sum to one; the oracle is omitted because it never sees the transformed data
Study 2: between-clip variability	Continuous indicators; clips differ in baseline and scale within the loading space	Number of dimensions and assignment of variables to them	DynEGA, static EGA, GraphicalVAR, GIMME, oracle	Whether recovery survives between-clip differences in selected conditions
Study 2: broad grid	Continuous indicators; all clips share one measurement model	Number of dimensions and assignment of variables to them	DynEGA	DynEGA recovery across a wider range of conditions

Method

The analyses follow the workflow in Figure 1. Table 1 summarizes what each analysis simulates, what it targets, and what may be concluded from it. Study 1 lets video clips differ in baseline and scale and produces bounded scores that sum to one, because that is what real classifier output looks like. Study 2 switches both off, so every clip shares one measurement model and the factor structure is the only source of shared variance. A sensitivity analysis then reintroduces between-clip differences in a restricted form. All three use the same LOMM machinery and differ only in what they license us to conclude.

For one simulated video clip, the implementation draws a starting state and the random acceleration inputs, moves each dimension forward with Equation 1, applies the activity schedule, draws the clip baseline and scale, evaluates Equation 2, and finally returns either the continuous indicators or the bounded scores of Equation 3 together with their log-ratio coordinates. The R function `lomm_observe()` performs the observation step and records which configuration was used, the random-number seeds, and the loading and covariance matrices in effect. A runnable example and the exact configuration behind the figures in this paper accompany the analysis code.

Study 1: Comparing Simulated and Real Score Series

Figure 3 illustrates the kind of real score series Study 1 compares against. MAFW, introduced as *MAFW: A Large-scale, Multi-modal, Compound Affective Database for Dynamic Facial Expression Recognition in the Wild*, supplies naturalistic video clips (Liu et al., 2022). The dataset does not validate the emotion labels, the scoring model, or the subjective emotional states of the people in the clips. Panel A reproduces four frames solely to clarify the academic analysis, following the small-portion publication allowance in the

MAFW Academic User Agreement.[1] No raw video is redistributed. Study 1 uses 100 video clips scored with the same zero-shot procedure as the illustration.

We used the transforEmotion R package to generate a score for each emotion label in each frame (Tomašević et al., 2026). The retained scores identify the `oai-large` model setting, which denotes the OpenAI CLIP ViT-Large/14 model family. The scoring procedure sampled up to 50 frames uniformly from each video. A Haar cascade detected faces in each frame, and the largest detected face was expanded by 50 pixels on each side before classification; frames without a detected face received missing scores. Eleven raw lowercase class labels—anger, disgust, fear, happiness, neutral, sadness, surprise, contempt, anxiety, helplessness, and disappointment—were passed directly to the text encoder, without a sentence template. These channels follow the MAFW single-label schema and remain model outputs rather than frame-level human annotations. Burned-in subtitles were not explicitly masked. Subtitle pixels could therefore contribute only when they fell inside the padded face crop. Observed scores were required to lie in $[0, 1]$; complete rows were retained only when their sum was positive, finite, and within 10^{-6} of one, then divided by that sum in the fixed component order. Values were not truncated, and zero replacement was disabled. The scoring-time package release and exact model-checkpoint revision were not archived, so exact frame rescoring cannot be guaranteed. The retained scores support all preprocessing and comparison analyses reported here. The main analysis kept only complete rows. A sensitivity analysis interpolated across internal gaps of at most two frames within a segment.

The four frames in Panel A come from one MAFW clip. Their scores differ across the channels labeled helplessness, fear, disgust, sadness, anxiety, and disappointment (Panel B), and the frame-by-frame trajectories change in bursts with irregular local movement (Panel C). Applying DynEGA to that single clip's non-neutral trajectories groups the channels into a network (Panel D). This last panel is an illustration of what these methods produce on real data, not evidence about what they recover; Study 2 addresses recovery using data whose structure is known.

The sample was assembled one video clip at a time, with MAFW category labels used to diversify it rather than as validation targets. The main analysis contains 3,557 frames in 188 clip segments; usable segment lengths range from 16 to 50 frames, with a median of 35. The gap-interpolated sensitivity analysis contains 3,632 frames in 129 segments, with lengths from 17 to 50 frames and a median of 36.5. Both retain 100 video clips spanning the 11 single-label categories. Because the segments are short, autocorrelation and spectral summaries should be read with caution.

[1] MAFW Academic User Agreement, accessed August 13, 2026: https://mafw-database.github.io/MAFW/academics/mafw-academics-final.pdf.

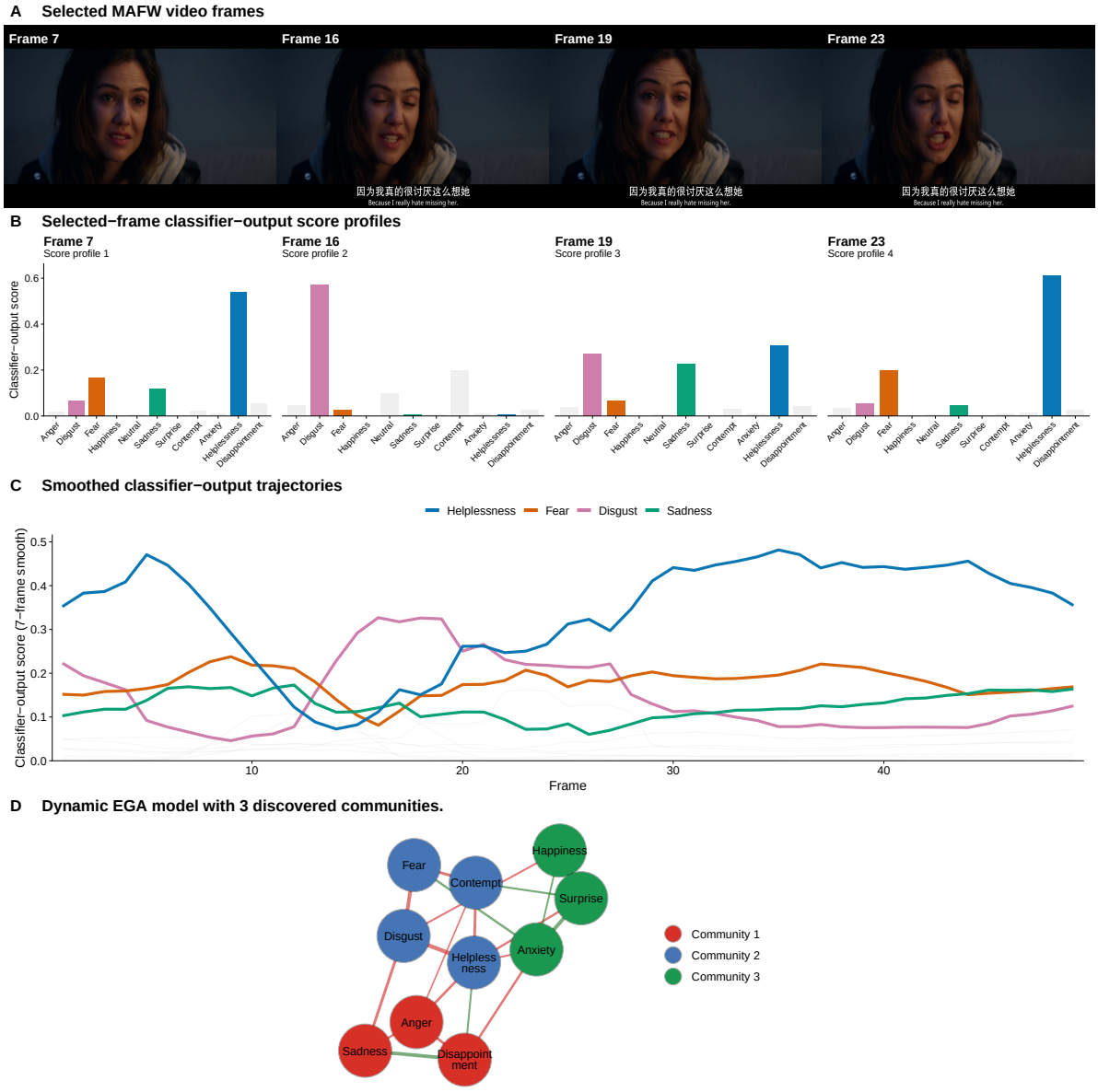


Figure 3: Score series from one MAFW video clip. Panel A shows four frames reproduced to clarify the academic analysis under the MAFW research-use terms. Panel B shows the emotion scores assigned to those frames; Panel C shows smoothed trajectories for the channels with the highest mean scores; Panel D shows a DynEGA network estimated from the same clip's non-neutral trajectories. Scores were generated with transforEmotion using OpenAI CLIP ViT-Large/14.

We compare simulated with real data on eight summary statistics. Six describe a single channel over time: the median level, the interquartile range of the level, the interquartile range of frame-to-frame changes, the lag-1 autocorrelation, the entropy of monotonic run lengths, and the normalized spectral entropy. Two describe all channels together: effective rank and total variance. Each statistic is computed twice, once on the scores themselves and once on their log-ratio coordinates, giving 16 comparisons in total. For each comparison, we take the larger of two distances between the real and simulated distributions of that statistic across clips: the absolute difference in medians and the one-dimensional Wasserstein distance, both divided by a measure of how spread out the real values are. That divisor is the interquartile range where available, falling back in order to $1.4826 \times$ MAD, the standard deviation, and one. Scores and log-ratio coordinates receive equal weight in the total, which we call the discrepancy. Smaller values mean the simulated data resemble the real data more closely on these particular statistics.

Video clips were split into five folds, assigned deterministically within MAFW category and length order, leaving 20 clips in each test fold. Everything about a fold's measurement model was estimated from its training clips only, so no test clip influenced the simulation it was later compared against. The center of the coordinate space was the mean of the training observations. After subtracting each clip's own mean, we kept the first three eigenvectors of the within-clip covariance, oriented each so that its largest coefficient was positive, and rescaled it by the square root of its eigenvalue divided by the root mean of the three retained eigenvalues. The covariance of the clip baselines was estimated from how far each training clip's mean fell from the overall mean, shrunk halfway toward its diagonal, with eigenvalues floored at 10^{-8} for numerical stability.

Within each fold and each configuration, we searched 27 candidate settings, crossing natural periods $\{8, 12, 24\}$, damping ratios $\{0.05, 0.20, 0.50\}$, and observation scales $\kappa \in \{0.75, 1.00, 1.25\}$, evaluating each on three paired training datasets. Selection averaged the eight discrepancies within each representation and weighted the two representations equally, breaking ties by the smaller worst-case discrepancy, then an observation scale closer to one, then stronger damping, then candidate order. All five folds selected the same settings for the configuration with both a clip baseline and a clip scale: a 12-observation period, a damping ratio of 0.50, and $\kappa = 1$. The tolerance below which the scaling divisor fell back to the next option was 10^{-8}.

Each selected configuration then generated 100 simulated datasets per test fold, matching the held-out clips' frame indices and segment structure so that the comparison is not confounded by differences in length or missingness. The discrepancy compares the distribution of each summary statistic across clips, not frame-by-frame values, since LOMM generates new realizations rather than reconstructing a particular video. The

composite weights the two representations equally and averages the five folds. For uncertainty, we drew 2,000 bootstrap samples, resampling 20 clips with replacement within each fold and preserving the pairing across configurations, datasets, and statistics. The eight configurations are the factorial combinations of three components: an estimated clip baseline, a clip scale, and a brief activity transition.

Choosing Simulation Scenarios

The scenarios connect Study 1 to Study 2, since both draw on the same latent and activity layers. Three sets of parameters define a scenario. The dynamic parameters are position feedback, velocity feedback, and acceleration-input variance (η_f, ζ_f, and σ^2_{qf}). The activity parameters determine whether one dimension dominates throughout a clip, whether dominance switches after sampled dwell intervals, or whether all dimensions stay near a low background level. The measurement parameters are the loadings and residual covariances that turn latent trajectories into observed indicators.

The design contains six scenario families (Table 2). Four are stable, so trajectories oscillate and decay. The remaining two use positive velocity feedback so that trajectories grow over the record; these are stress tests rather than empirically calibrated affective regimes. Across families, the number of observed channels is $p = mk$, where m is the number of dimensions and k the number of indicators per dimension. Study 2's main benchmark uses $m \in \{2, 3\}$, and the broad grid adds $m = 4$. This range defines a deliberately low-dimensional recovery target; it is not an estimate of how many dimensions underlie real classifier scores, a question Study 1 addresses only indirectly through the effective-rank statistic.

Table 2: The six scenario families. Dwell values are fractions of the time-series length; the codes label conditions in the results figures.

Code	Family	Switching	Active/background gain	Dwell fraction	Dynamic rule
A	Weak expression, one dominant dimension	No	0.35 / 0.05	1.00 / 1.00	Stable; low input scale
B	Strong expression, one dominant dimension	No	1.00 / 0.08	1.00 / 1.00	Stable; standard input scale
C	Strong expression, switching dimension	Yes	1.00 / 0.08	0.20 / 0.45	Stable; standard input scale
D	Strong expression, rapid switching	Yes	1.25 / 0.08	0.03 / 0.10	Stable; high input scale
E	Growing trajectories, one dominant dimension	No	1.00 / 0.08	1.00 / 1.00	Positive velocity feedback; moderate input scale
F	Growing trajectories, switching dimension	Yes	1.10 / 0.10	0.12 / 0.30	Positive velocity feedback; higher input scale

Appendix F lists the exact feedback and acceleration-input values behind each family.

For each video, LOMM generates the latent trajectories, applies the activity schedule, and then constructs the observed data. Non-switching scenarios keep one dimension dominant for the whole clip and rotate which dimension that is across videos. Switching scenarios sample dwell intervals and change the dominant dimension over time. Study 1 uses bounded scores with a clip baseline and clip scale. Study 2 uses continuous indicators with uncorrelated measurement error and the small loading and variance perturbations described above.

Study 2: Recovery of Dimensions from Data with Known Structure

Study 2 starts from a known answer: the number of dimensions and the assignment of each indicator to a dimension are both fixed before any data are generated. Its main conditions give every video clip the same measurement model, and the sensitivity analysis adds between-clip differences that stay within the space spanned by the loadings. Bounded scores are reserved for Study 1, because constraining scores to sum to one changes the associations among the original channels and therefore no longer preserves the structure we planted.

The four methods estimate different things, so they are compared at the one point where their outputs are commensurable: the grouping of variables. Each method receives the same simulated data, estimates its own structure, converts that structure into a network by a rule fixed in advance, and then passes the network to the same community-detection algorithm. What that algorithm returns is compared with the true assignment of variables to dimensions. The methods do not all use the same rows of data on the way there — DynEGA loses rows to time-delay embedding, GraphicalVAR loses rows to forming lagged pairs — and we deliberately do not force them onto a common subset, since each method's row handling is part of the method. Two things this comparison does not address: whether the emotion labels attached to real channels are valid, and which method is better for substantive emotion research.

We compare Dynamic Exploratory Graph Analysis [DynEGA; H. Golino et al. (2022)] with GraphicalVAR (Epskamp et al., 2018; Epskamp, 2024) and GIMME (Gates & Molenaar, 2012; Lane et al., 2019) because they are neighboring approaches to structure in intensive longitudinal data. DynEGA is the focal method because it estimates groups of variables whose changes are coordinated over time, which is directly what Study 2 asks about. GraphicalVAR separates relations across consecutive time points from relations within the same time point; we group the variables using the latter. GIMME estimates a separate model for each video clip and we aggregate the resulting structures before grouping, which places it under more strain than the other methods and makes it as much a test of feasibility as of accuracy. Treating clips as separate series in this way is not how GIMME is normally used, and our results should not be read as evidence about its

performance in person-level idiographic modeling.

DynEGA combines time-delay embedding, derivative estimation, and exploratory graph analysis (Boker et al., 2010; H. Golino et al., 2020; H. F. Golino & Epskamp, 2017). Each video is passed through the `id` argument so that embedding windows never span two clips. We selected the embedding dimension with an explicit loop that fit every admissible candidate and minimized the Total Entropy Fit Index [TEFI; H. Golino et al. (2021)], breaking ties in favor of the shorter embedding. For $T = 100$ the candidates were $\{5, 10, 15, 25, 50\}$; for the shorter series we used $\{5, 6, 10, 12\}$. Selected counts for $T = 100$ were 141, 242, 7,206, 5,509, and 8,502, respectively; for $T = 25$ they were 2,327, 2,735, 6,068, and 10,470. EGAnet 2.3.0 estimated first derivatives with `tau = 1`, a `delta = 1` observation interval, and `use.derivatives = 1`, retaining the default `corr = "auto"` and `na.data = "pairwise"` settings. Neither our code nor the executed package path added noise to zero-variance channels. Rescaling every derivative from observation units to physical-time units would multiply all channels by the same positive constant, leaving the correlation matrix unchanged. We set `model = "glasso"` to estimate a regularized partial-correlation network with EBICglasso. Before community detection, each fitted network was symmetrized and its negative edges set to zero, with absolute weights used only when no positive edge remained. The seed for each simulated dataset determines its random draws, and a deterministic method-specific offset initializes each fit. Louvain community detection at resolution 1.0 was the primary grouping algorithm (Blondel et al., 2008); Leiden (Traag et al., 2019) and Walktrap (Pons & Latapy, 2005) were applied to the same fitted network as sensitivity analyses.

GraphicalVAR estimates a graphical vector autoregressive model for intensive longitudinal data (Epskamp et al., 2018; Epskamp, 2024). It produces two networks: a temporal network of directed regressions from one time point to the next, and a contemporaneous network of partial correlations among residuals at the same time point, after the lagged effects have been accounted for. In multilevel applications it can further separate within-person from between-person structure. We fit it with `mlGraphicalVAR` to the stacked simulated data, with video identity as the grouping variable, within-clip centering (`centerWithin = TRUE`), variable standardization (`scale = TRUE`), and `gamma = 0.5`. Variables are standardized across the stacked data and then centered within clip before lagged pairs are formed, and those pairs never cross a clip boundary. We group the variables using the returned contemporaneous partial-correlation matrix `fixedPCC`, because Study 2 asks about an undirected grouping of channels rather than about directed effects. Its diagonal is ignored and the result is treated as an undirected weighted graph defined by positive edges, with absolute weights used only when no positive edge is available. Louvain, Leiden, and Walktrap are applied to that same fitted structure.

GIMME is an automated time-series approach built on unified structural equation modeling and vector

autoregression (Gates & Molenaar, 2012; Lane et al., 2019). Each video clip was supplied as one time series with standardized variables. We extracted the standardized residual covariance matrices returned in `fit$psi`, then averaged them over usable clips, symmetrized them, removed the diagonal, and set negative values to zero before community detection. The resulting graph is a residual-covariance graph, not a residual-correlation or partial-correlation graph: it keeps the magnitude of residual covariance as the edge weight, so unequal residual variances can influence the grouping. We therefore read GIMME's results as specific to this conversion. Variable names fixed the column order across clips. A matrix left with no positive edge yielded one community per variable rather than falling back to absolute weights. Louvain was primary; Leiden and Walktrap were applied to the same graph.

DynEGA, static EGA, GraphicalVAR, GIMME, and the oracle all receive the same simulated data and are judged against the same true grouping. GraphicalVAR and GIMME were left out of the broad 518,400-dataset extension because evaluating them across a complete factorial design was not computationally practical; both were included in the main benchmark, and GraphicalVAR was also included in the targeted departures and the between-clip-variability analysis. Appendix B places the networks these methods estimate on a common population scale.

The main $T = 100$ design contains 21,600 simulated datasets: six scenario families, $m \in \{2, 3\}$ dimensions, $N \in \{50, 100\}$ videos, residual standard deviations of $\{0.125, 0.25, 0.50\}$, loadings of $\{0.40, 0.60, 0.80\}$, four indicators per dimension, and 100 paired replications. Every method received the same dataset within each condition and replication. Louvain at resolution 1.0 was the primary grouping algorithm, with Leiden and Walktrap as paired sensitivity analyses.

Static EGA applies EBICglasso and the same grouping step to each dataset's contemporaneous correlation matrix, ignoring the time ordering entirely. The oracle skips estimation and simply passes a graph containing the true within-dimension edges, and no others, through that same grouping step. It therefore verifies that the conversion, variable ordering, and scoring work correctly, and provides a ceiling that no method estimating from data could be expected to exceed.

Not every fit succeeds, and how failures are counted turns out to matter a great deal for the comparison. We therefore report three quantities. The first is the probability that a method returned a usable result at all, $P(\text{usable})$. The second is the probability that it found the correct number of dimensions given that it did, $P(\widehat{m} = m \mid \text{usable})$. The third combines them into the probability that a method both produced a usable result and got the number of dimensions right, $P(\text{usable and correct})$. Because this last quantity counts every failure and timeout as an error, we call it *worst-case accuracy*, and it is the headline number

throughout: a method that often fails to return anything is not useful even if it is accurate whenever it does.

Three further diagnostics describe the grouping itself, among usable fits only. The adjusted Rand index (ARI) and variation of information compare the estimated grouping with the true one. Assignment accuracy matches estimated groups to true dimensions optimally, counts every indicator in the denominator, and treats an indicator placed in an unmatched group as incorrect. We also compare recovered trajectories with the activity-scaled latent positions $\tilde{x}$ that actually entered the measurement model. Within each dataset, indicators are standardized and each estimated group is summarized by its first principal component, falling back to a row mean when that cannot be computed. Trajectory correlation pairs each true dimension with its highest absolute correlation among the estimated summaries, which neither penalizes extra groups nor requires one-to-one matching. NRMSE instead matches estimated with true trajectories one-to-one after sign alignment, assigning the prespecified penalty of 2 to any true dimension left unmatched while ignoring extra estimated groups. In the marginal summaries, unusable fits receive a correlation of 0 and an NRMSE of 2. None of these diagnostics replaces worst-case accuracy.

Errors and timeouts stay in the denominator of every worst-case rate. Because the three community-detection algorithms are applied to one fitted structure rather than to three separate fits, they are sensitivity analyses rather than independent observations, and the main text reports Louvain only. Uncertainty comes from 2,000 resamples of whole simulated datasets, stratified by condition, with Monte Carlo standard errors from delete-one-dataset jackknifing.

A second analysis repeated the same 21,600-dataset design with shorter time series ($T = 25$, $\Delta = 0.4$, spanning 9.6 units of physical time). A set of targeted departures then fixed $N = 100$, $T = 100$, a loading of 0.60, a residual standard deviation of 0.25, and four indicators per dimension, crossing the six scenario families, two dimensionalities, and 50 paired replications; Table 3 states exactly what each departure changes. The broad DynEGA extension crossed $N \in \{50, 100\}$, $T \in \{100, 300\}$, $m \in \{2, 3, 4\}$, $k \in \{4, 6\}$, loadings of $\{0.40, 0.60, 0.80\}$, residual standard deviations of $\{0.125, 0.25, 0.50\}$, and the six scenarios, giving 1,296 conditions with 400 replications each (518,400 datasets). It used the EGA package's own rule for deciding whether a fitted network was unidimensional. Because $T = 100$ and $T = 300$ use $\Delta = 0.4$ and $\Delta = 0.25$ respectively while holding the per-step input variance fixed, the difference between them reflects both the number of observations and the interval between them, not length alone.

Table 3: The targeted departures. Each row changes one element of the ideal condition, holding everything else fixed.

Departure	Specification
Independent starting states	Each video and dimension starts independently: stationary zero-mean starts for stable dimensions, $x_0 \sim \text{Unif}(-0.25, 0.25)$ and $v_0 \sim \text{Unif}(-2, 2)$ for growing ones
Gradual activity transitions	A five-observation gradual transition replaces each abrupt switch
Cross-loadings	Each indicator gains small loadings on other dimensions: $N(0, 0.05^2)$ additions truncated to $[-0.10, 0.10]$
Correlated residuals within a dimension	Residual correlation of 0.10 among the indicators of each dimension
Correlated inputs across dimensions	Acceleration-input correlation of 0.15 between every pair of dimensions
Bounded scores	Scores mapped through the softmax with $\tau = 1$, analyzed either as scores or as log-ratio coordinates

Computational Environment

Analyses used R 4.5.1 on Rocky Linux 8.8 (`x86_64-redhat-linux-gnu`) with dual Intel Xeon Silver 4310 processors, 48 logical threads, and 125 GiB RAM. Key package versions were EGAnet 2.3.0, graphicalVAR 0.3.4, gimme 10.0, igraph 2.2.1, qgraph 1.9.8, psych 2.5.6, lavaan 0.6-20, dplyr 1.1.4, and ggplot2 4.0.1. OpenBLAS 0.3.15 and LAPACK 3.9.0 supplied numerical linear algebra. Parallel analyses used one BLAS thread per worker. DynEGA used 16 workers on the broad grid and 32 for the matched-network partitions; GraphicalVAR and GIMME used 24 workers with a 900-second limit per structure fit.

Runtime is elapsed wall-clock time in seconds, recorded with `proc.time()[["elapsed"]]` for fits that succeeded. For GraphicalVAR and GIMME it includes both fitting the structure and converting it into a grouping. Because one fitted structure is reused for Louvain, Leiden, and Walktrap, we count fits and results separately: feasibility is reported per unique fit, whereas accuracy is reported per fit-and-algorithm combination. Runtime summaries use successful fits only, while timeouts and errors remain in the denominators for feasibility and accuracy. These runtimes describe what this workflow costs on this machine and should not be read as hardware-independent speed benchmarks for the underlying methods.

Results

Study 1 Results: Similarity of Simulated and Real Score Series

Study 1 asks whether simulated score series reproduce chosen statistical properties of real classifier scores from 100 MAFW video clips. We name the eight configurations by which components they include. In the *identical-clips* configuration, every video shares one measurement model. The *baseline-only* configuration lets clips differ in their overall score level. The *baseline-and-scale* configuration lets them differ in both level and variability. The remaining four configurations add a brief activity transition to each of these. Figure 4 reports the mean discrepancy for all eight, computed on held-out clips and shown separately for the scores themselves and for their log-ratio coordinates. Smaller values mean closer agreement.

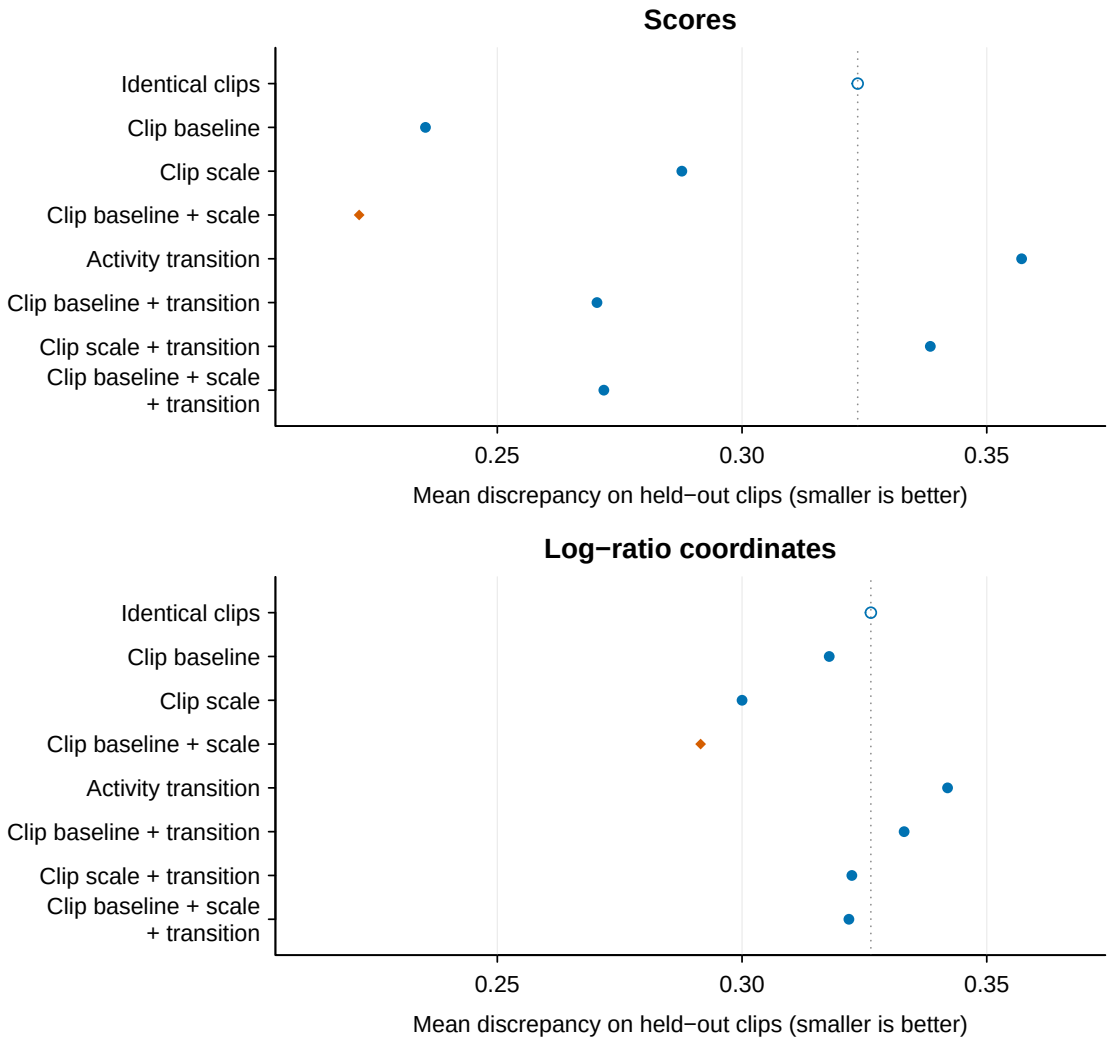


Figure 4: Discrepancy between simulated and real score series for the eight configurations, averaged over five cross-validation folds. Points are means across the eight summary statistics within each representation; smaller values indicate closer agreement with the real scores from 100 MAFW video clips. The diamond marks the baseline-and-scale configuration. Bootstrap intervals are not shown for individual configurations because their distributions were strongly displaced from the estimates themselves; Appendix C reports those diagnostics.

Averaging over both representations, the baseline-and-scale configuration had a mean discrepancy of 0.2567, compared with 0.3250 for the identical-clips configuration, a reduction of 21.0%. Letting video clips differ from one another therefore brought the simulated data substantially closer to the real scores. The bootstrap reproduced each estimate exactly when every clip received unit weight, but its resampled distributions sat well above the estimates themselves, as often happens with a distance that cannot go below zero and is not a smooth function of the data. For the baseline-and-scale configuration on the score scale, the estimate was 0.2218 while the bootstrap mean was 0.2701, with a basic interval of $[0.1433, 0.1992]$ and a percentile

interval of $[0.2444, 0.3002]$. Since neither interval brackets the estimate in a useful way, Figure 4 reports the means without intervals, and Appendix C retains the bootstrap means, bias estimates, and both interval constructions as diagnostics.

Taking the two components separately, adding a clip baseline changed the discrepancy by -0.0418 and adding a clip scale by -0.0188, where negative values favor including the component. Their bootstrap means were -0.0343 and -0.0072, implying biases of 0.0074 and 0.0116. The basic intervals were $[-0.0577, -0.0423]$ and $[-0.0385, -0.0233]$, and the percentile intervals were $[-0.0413, -0.0258]$ and $[-0.0143, 0.0009]$. The clip-scale interval includes zero under one construction and not the other, so we read these paired comparisons descriptively and make no claim of statistical significance. The baseline is the component doing most of the work; the scale helps less and remains optional. Adding the brief activity transition made the discrepancy worse.

The sensitivity analysis that interpolated short gaps produced the same ordering. There the baseline-and-scale configuration had a discrepancy of 0.2511, against 0.3187 for identical clips and 0.2707 for baseline-only, reductions of 21.2% and 7.3%. The baseline-and-scale configuration had the lowest discrepancy of all eight in both representations and both analyses. The individual statistics also show where the resemblance breaks down: in the main analysis the worst agreement was 0.3499 for total variance on the score scale and 0.4423 for the interquartile range of frame-to-frame changes in log-ratio coordinates, the latter rising to 0.4639 after interpolation. Simulated series are therefore still too smooth or too rough from frame to frame in ways the current configuration does not capture, and Appendix C reports every configuration and every statistic. One further caveat bounds all of this: the same test folds were used both to compare the eight configurations and to pick the best one, so the comparison is exploratory and does not account for the uncertainty in that choice. An unbiased assessment would need a new dataset or an outer validation split. Within that limit, the results support the baseline-and-scale configuration as a reasonable starting point for this kind of score series while leaving clear targets for later work. They say nothing about whether the scoring model or its emotion labels are valid.

Figure 5 shows what this resemblance looks like in a single case. We fixed the 49-frame MAFW clip from Figure 3 and generated 100 realizations with the baseline-and-scale configuration. The one displayed is the closest of the 100, chosen to minimize the equally weighted mean of the root mean squared error in the scores and in their frame-to-frame changes, each standardized by the corresponding scale in the real data, across all 11 emotions. Its score RMSE was 0.105 and its first-difference RMSE was 0.083.

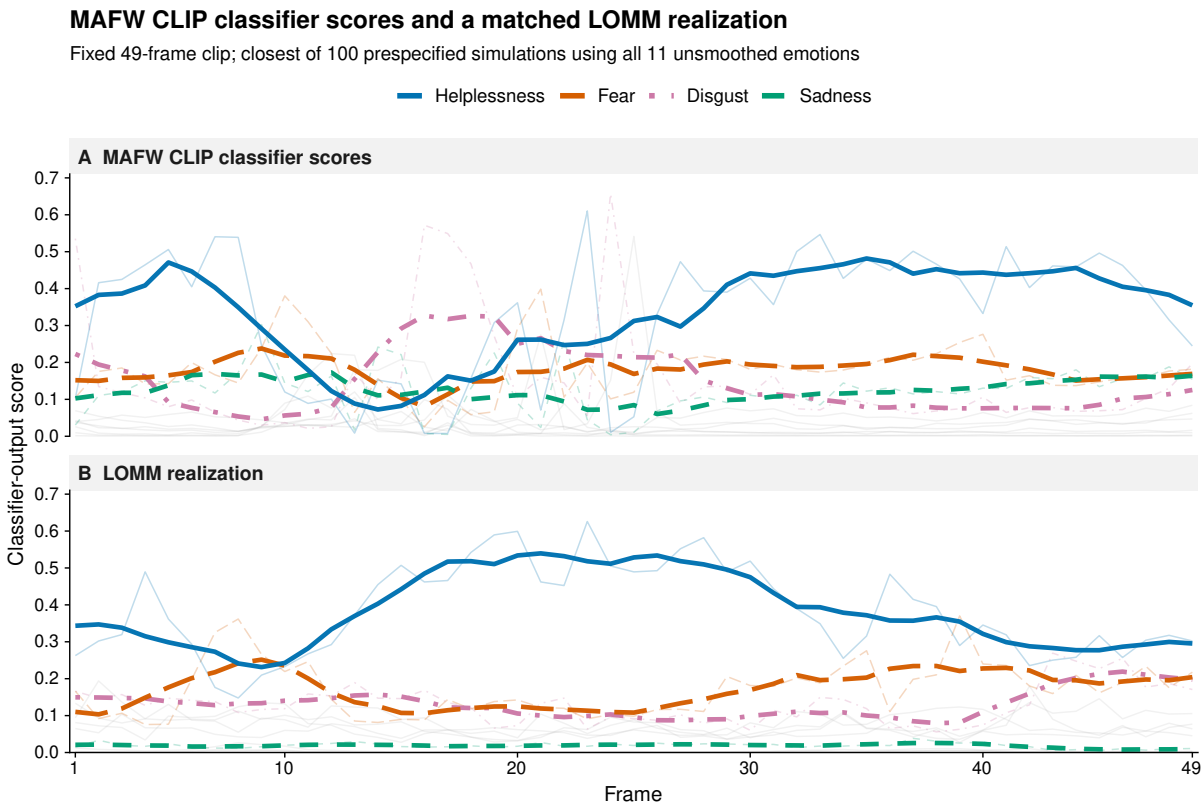


Figure 5: Real MAFW classifier scores compared with the closest of 100 LOMM realizations generated for the same 49-frame clip, using the baseline-and-scale configuration. The realization shown minimized the equally weighted mean of the root mean squared error in the scores and in their frame-to-frame changes, each standardized by the corresponding scale in the real data, across all 11 emotions. Thick lines show seven-frame moving averages for the four emotions with the highest mean real scores; faint lines show the unsmoothed scores and the remaining emotions. Because this realization was selected after the fact, it illustrates the best resemblance attainable rather than typical agreement; Figure 4 carries the aggregate evidence.

The selected realization matches the real series in the general size of its scores, in how much they move across several frames, and in the way higher- and lower-scoring emotions vary together. Its peaks do not fall on the same frames, and they should not: LOMM generates new random realizations rather than reconstructing a particular video. Because this series was picked after the fact as the closest of 100, Figure 5 shows a best case rather than typical agreement; the five-fold results above are the aggregate evidence.

Study 2 Results: Recovery of the Number of Dimensions

Table 4 reports the Louvain results for the main $T = 100$ benchmark. Recall that worst-case accuracy requires both a usable fit and the correct number of dimensions, so it charges a method for every failure and timeout. Static EGA and the oracle appear alongside the three methods that use the time ordering.

Table 4: Louvain results for the main benchmark, with 100 observations per video clip. Each simulated dataset is one independent observation. Usability and worst-case accuracy use all datasets; accuracy conditional on usability uses only the fits that succeeded. Parentheses contain Monte Carlo standard errors.

Method	Fits	Usable	Usable and correct	P(usable)	P(correct | usable)	P(usable and correct)
DynEGA	21600	21600	20282	1.000 (0.0000)	0.939 (0.0016)	0.939 (0.0016)
GraphicalVAR	21600	20329	16784	0.941 (0.0016)	0.826 (0.0027)	0.777 (0.0028)
GIMME	21600	4846	3795	0.224 (0.0028)	0.783 (0.0059)	0.176 (0.0026)
Static EGA	21600	21526	19093	0.997 (0.0004)	0.887 (0.0022)	0.884 (0.0022)
Block oracle	21600	21600	21600	1.000 (0.0000)	1.000 (0.0000)	1.000 (0.0000)

All 21,600 DynEGA fits were usable, and 93.9% returned the correct number of dimensions. Static EGA reached 88.4% worst-case accuracy, which shows that a great deal of the structure was already visible in the score levels without using the time ordering at all. GraphicalVAR produced usable fits for 94.1% of datasets and 77.7% worst-case accuracy. GIMME was accurate when it worked, getting the number of dimensions right in 78.3% of its usable fits, but it produced a usable fit only 22.4% of the time, which brings its worst-case accuracy down to 17.6%. The oracle recovered every grouping, confirming that the shared machinery was not itself a source of error.

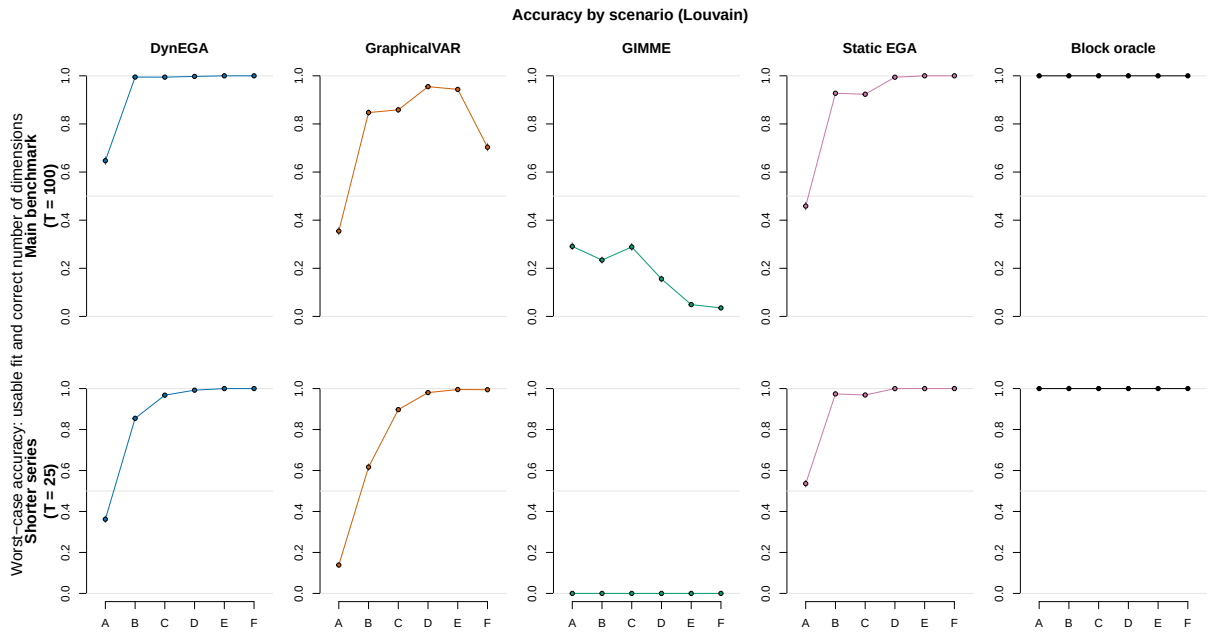


Figure 6: Worst-case accuracy under Louvain across the six scenario families, with 25 or 100 observations per video clip. Intervals are 95% bootstrap intervals resampling whole datasets. Condition codes are defined in Table 2.

Figure 6 shows that accuracy varied both by scenario and by the length of the time series. DynEGA was the strongest method in most conditions. Static EGA's high overall accuracy again reflects how much

structure was available in the score levels alone when every video clip shared one measurement model; the targeted departures below, particularly independent starting states and between-clip variability, separate the two methods more clearly. GraphicalVAR struggled when expression was weak or measurement was poor. GIMME's low numbers are driven almost entirely by fits that never returned a usable result.

DynEGA also completed the broad grid of 518,400 datasets. Using its own grouping rule, it recovered the correct number of dimensions in 93.7% of datasets and assigned individual indicators to the right dimension 95.2% of the time. Accuracy generally improved with stronger loadings and smaller residual variance, as expected. The $T = 300$ conditions also changed the interval between observations, so the difference from $T = 100$ reflects the whole design rather than length alone. A separate diagnostic run on the generator characterized the two growing-trajectory families in representative conditions ($N = 50$, $k = 4$, loading 0.60, residual SD 0.25; 25 replications). The median largest absolute latent value ranged from 55.9 to 105.2 at $T = 100$ and from 329.4 to 1,201.6 at $T = 300$, and the final tenth of each series accounted for between 0.316 and 0.517 of the total latent energy, with no infinite or missing values. Late-record amplitude can therefore dominate the covariance in those conditions, which is why we treat them as stress tests (Appendix F).

Leiden and Walktrap gave almost the same accuracy as Louvain. The largest spread among the three algorithms was 0.0059 at $T = 100$ and 0.0033 at $T = 25$, and the targeted departures were similarly stable, with a largest spread of 0.045 for DynEGA on bounded scores and no change in the overall pattern. Appendix G reports these comparisons.

Appendix D reports how well each method recovered the grouping and the trajectories among its usable fits, using ARI, assignment accuracy, trajectory correlation, and NRMSE. Those numbers describe what a method achieves when it works and should be read alongside, not instead of, the worst-case accuracy in Table 4.

Robustness and Sensitivity Analyses

The remaining analyses answer the third question, which concerns how far these results survive departures from ideal conditions. Shortening the time series to $T = 25$ gave worst-case accuracy of 0.863 for DynEGA, 0.913 for static EGA, 0.770 for GraphicalVAR, 0 for GIMME, and 1.000 for the oracle. Compared with $T = 100$, shorter series cost DynEGA a little, left GraphicalVAR roughly where it was, and made GIMME entirely infeasible. Static EGA actually did somewhat better on the shorter series.

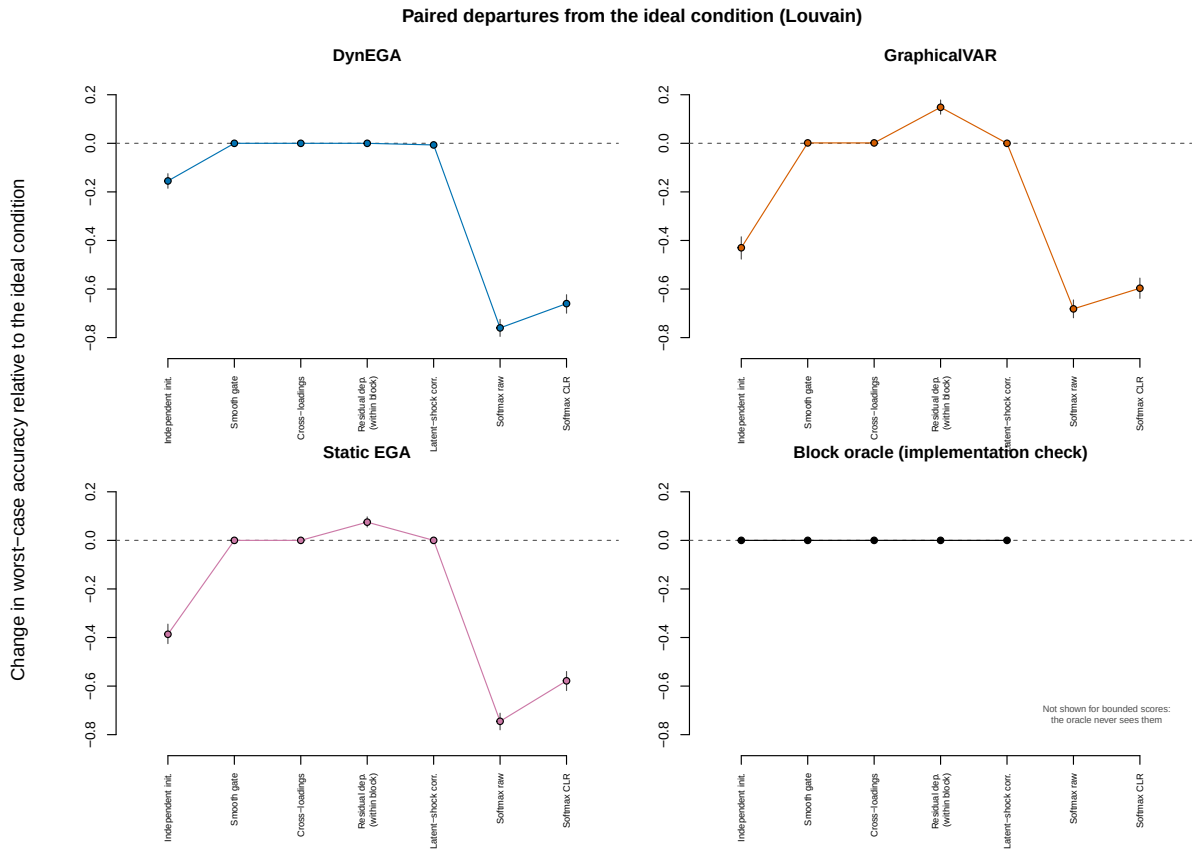


Figure 7: Paired change in worst-case accuracy under each targeted departure, relative to the ideal condition. DynEGA, static EGA, and GraphicalVAR appear throughout; the oracle appears only for the continuous-indicator departures, and GIMME was not fitted here. The oracle is omitted for bounded scores and log-ratio coordinates because it receives the true structure directly and never sees the transformed data. Negative values mean worse accuracy. Intervals are 95% bootstrap intervals resampling whole datasets.

Independent starting states were by far the most consequential departure. Accuracy fell from 0.993 to 0.838 for DynEGA, from 0.912 to 0.525 for static EGA, and from 0.683 to 0.253 for GraphicalVAR, while the oracle stayed at 1.000. Giving every dimension the same starting state had therefore been making the main benchmark easier than it looks, a point we return to in the Discussion and that qualifies the population argument in Appendix B. Replacing abrupt switches with five-observation gradual transitions changed almost nothing in the conditions we tested: DynEGA stayed at 0.993, static EGA at 0.912, and GraphicalVAR moved from 0.683 to 0.685. Cross-loadings and correlated inputs across dimensions also produced small changes. Correlated residuals within a dimension actually helped GraphicalVAR and static EGA here, which makes sense: unlike the other two departures it adds no association between dimensions and so reinforces the very grouping the methods are looking for. These are one-at-a-time changes and should not be read as a general ranking of which conditions are hardest.

Bounding the scores changed the question being asked and sharply reduced accuracy on the original channels. DynEGA recovered the correct number of dimensions in 0.333 of the log-ratio datasets and 0.233 of the bounded-score datasets; the corresponding figures were 0.333 and 0.167 for static EGA, and 0.087 and 0.002 for GraphicalVAR. The oracle is omitted here because it receives the true structure directly and never sees the transformed data, so its unchanged result would tell us only that the grouping step still works, not whether constraining scores to sum to one preserves the structure among the original channels. It does not, and these results are the reason Study 1 and Study 2 have to stay separate.

A final analysis added between-clip differences in baseline and scale to selected stable $T = 100$ conditions, drawing the baselines within the space spanned by the loadings. It crossed scenarios A–D, $m \in \{2, 3\}$, and 10 paired replications, giving 80 paired datasets, each with $N = 100$, $k = 4$, a loading of 0.60, a residual standard deviation of 0.25, and independent stationary starts; GraphicalVAR and GIMME had fitting limits of 300 and 600 seconds. DynEGA's worst-case accuracy was 0.750 with between-clip differences and 0.775 without, a paired difference of -0.025 with a 95% percentile interval of $[-0.100, 0.050]$. Static EGA rose from 0.275 to 1.000, because a baseline lying in the loading space adds strong between-clip signal pointing at exactly the right grouping, and static EGA is well placed to use it. GraphicalVAR stayed at 0.0375 and GIMME at 0.400 under both, and the oracle stayed at 1.000. GraphicalVAR's figure rests on three correct fits out of 80 deliberately chosen datasets and is not comparable with its 0.777 across the main benchmark. This was a focused compatibility check rather than a test of equivalence: DynEGA's interval still permits a loss as large as 0.10, every three-dimensional GIMME fit timed out, and GraphicalVAR's unchanged accuracy came with modest declines in ARI and assignment accuracy.

Computational Feasibility

Figure 8 summarizes how long successful fits took. Counts refer to unique fitted structures, before any structure is reused for a second community-detection algorithm.

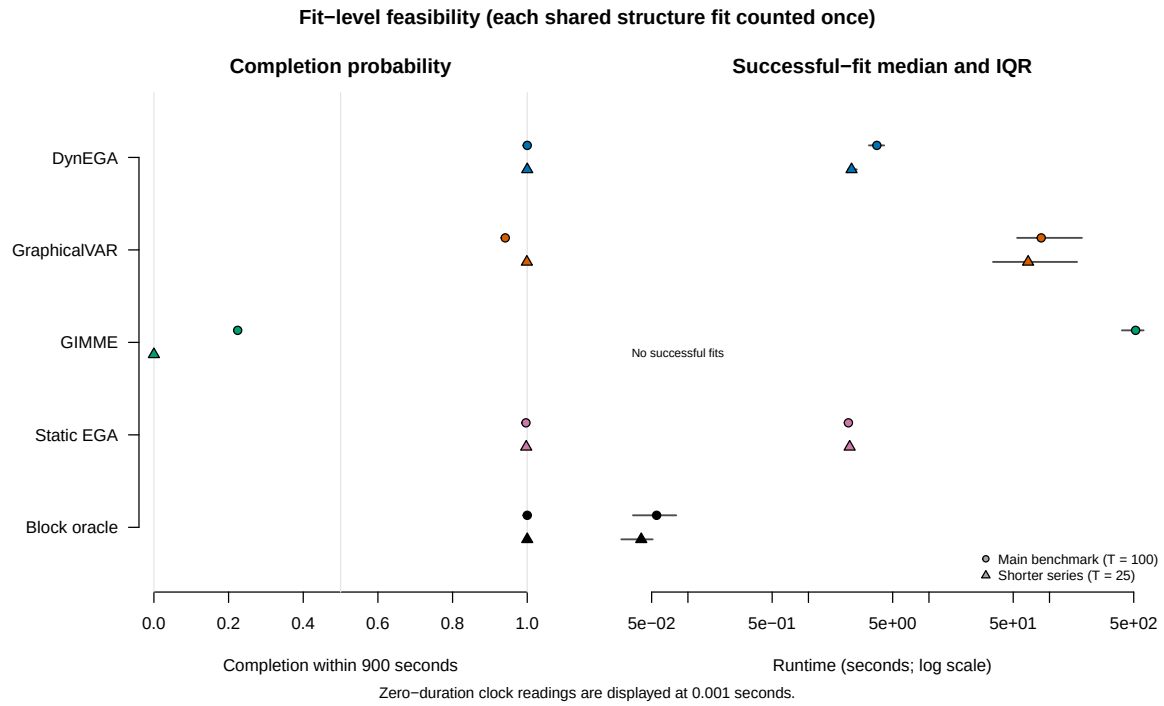


Figure 8: Runtime of successful fits in the main benchmark. Points show medians and intervals show interquartile ranges on a log10 scale. Counts use unique fitted structures.

Median runtime for a successful fit was 3.69 s for DynEGA, 85.41 s for GraphicalVAR, and 518.16 s for GIMME, a difference of more than two orders of magnitude between the fastest and slowest. GraphicalVAR completed 20,329 of 21,600 fits and timed out on 1,271. GIMME completed 4,846, timed out on 14,477, and returned errors after estimation for a further 2,277. DynEGA completed every fit in both the main benchmark and the broad grid. These figures describe the specific implementations on this machine and are not a general ranking of how fast these families of methods are. Appendix E gives the full accounting of successes, timeouts, and errors.

Discussion

LOMM addresses a problem that is becoming common in behavioral research. A classifier will happily produce a dense stream of scores from any video, and a dynamic method will happily return a network or a set of dimensions from that stream, but nothing in the data tells us whether those dimensions were there to be found. LOMM makes that question answerable by separating the latent dynamics, the pattern of expression over time, and the measurement model that produces observed scores, so that each can be changed while the others are held fixed. That separation is the paper's central contribution, because it lets

a result be traced back to the part of the generating process responsible for it.

The two studies serve complementary purposes and neither substitutes for the other. Letting video clips differ in baseline and scale brought the simulated scores substantially closer to the real MAFW scores on the summary statistics we chose. That finding is local to one dataset, one set of prompts, and one scoring model, and because the same test folds were used both to compare configurations and to select the best one, it is exploratory rather than an unbiased external validation. It establishes resemblance on chosen statistics; it does not establish that the emotion labels are valid or that LOMM describes how emotional expression actually works. The clip baseline was the component that mattered; the clip scale helped less and remains optional. Reporting all eight configurations in Figure 4 keeps that choice visible.

Study 2 then supplied a structure known in advance. DynEGA had the highest worst-case accuracy of the methods that estimate from data and was the only one that could complete the broad grid. Static EGA also performed well, which shows that a good deal of the structure was recoverable from score levels alone, without using the time ordering. The two methods came apart under the departures. DynEGA barely changed when video clips were allowed to differ in baseline and scale, whereas static EGA improved sharply, because those differences lie in exactly the space it is looking at. Both lost accuracy under independent starting states. GraphicalVAR often recovered the right number of dimensions when it produced a usable fit, but it was more sensitive to conditions and far more expensive to run. GIMME was usually right when it worked and usually did not work. All of this describes the specific implementations and network conversions we used.

The independent-starting-state result deserves particular emphasis, because it qualifies the main benchmark. Starting every dimension from the same state creates a transient that all dimensions share, which is structure that has nothing to do with the factor model but is nonetheless recoverable. Removing it reduced accuracy for DynEGA, static EGA, and GraphicalVAR alike. Initialization is therefore not an incidental implementation detail but part of the data-generating design, and simulation studies that hold it fixed may be reporting easier problems than they intend. By contrast, smoothing the activity transitions changed little. That comparison concerns the accuracy rate only; it does not separate how much of the derivative signal comes from the oscillator, from the transition, and from measurement error.

Appendix B offers a more general insight into methods that work through networks. Under an idealized model with equal loadings and equal error variances, both the derivative-based and the lagged-residual network preserve the true grouping, and both convert a signal-to-noise ratio into the strength of a within-dimension edge. Crucially, their partial correlations share the same upper bound of $1/(k-1)$. That bound

holds regardless of which network is used, so building a network from derivatives carries no automatic advantage at the population level. Whatever differences we observe must come from the signal-to-noise ratio each method achieves, from the length of the time series, from regularization, from how the network is constructed, and from whether the method runs at all. The bound is a mathematical result, not an estimate of why the methods differed in our data. Its wider lesson is that adding common signal cannot strengthen a partial correlation without limit when many indicators are conditioning on one another.

Two details keep that argument aligned with what we actually ran. GraphicalVAR centers within each clip, which slightly reduces residual variance and introduces a small negative covariance between adjacent errors, of order $1/T$; Appendix B carries those terms through, and they change the signal-to-noise ratio without changing the structure of the result. For GIMME, the network we evaluate is built from standardized residual covariances after fitting, which is neither a residual-correlation nor a directed-path network.

The bounded-score results mark a firm boundary on what we can claim. Constraining scores to sum to one changes the associations among the original channels, and neither the scores nor their log-ratio coordinates preserved the result obtained with continuous indicators. A study of recovery from bounded scores would need its own definition of the truth, such as a grouping specified directly in log-ratio coordinates with a fixed basis. Study 1 shows that the generator resembles real scores on selected features; Study 2 shows that the methods recover a known structure from continuous indicators. Neither statement licenses the other.

Several limitations follow from the design. The latent dimensions are uncoupled, and allowing them to drive one another could change both the population argument and the ranking of methods. The growing-trajectory scenarios produce very large amplitudes late in the series and are stress tests rather than plausible affective regimes. The $T = 300$ conditions change both the number of observations and the interval between them, so they do not isolate length. GIMME's residual-covariance conversion was built for this comparison and does not represent how the method is normally used. Finally, the real clips are short and come from a single controlled-access dataset scored by a single zero-shot pipeline.

For applied methods work, LOMM suggests a practical order of operations. Decide what the measurement actually is, calibrate only the observable features that matter for the application at hand, specify a truth appropriate to that measurement, and only then evaluate what methods recover and whether they run. Working in that order makes it harder for a classifier's output to acquire substantive meaning simply because a dynamic method returned something interpretable. The present results support LOMM as an environment for that kind of validation, and they point toward extensions with other scoring systems and datasets, coupled rather than independent dimensions, and truth targets defined directly for bounded scores.

Declarations

Funding

The authors received no specific funding for this work.

Competing interests

The authors declare that they have no competing interests.

Ethics approval

This study used simulations and secondary video data obtained under the MAFW research-use terms. It involved no new recruitment, intervention, or direct collection of human-participant data by the authors.

Consent to participate

Not applicable.

Consent for publication

The manuscript includes four still frames from MAFW solely to illustrate the score-extraction procedure. The frames are cited to the dataset, and no source video is redistributed. Their inclusion follows the dataset's terms for limited reproduction in academic publications.

Availability of data and materials

The simulation results, derived classifier-output score summaries, and analysis-ready data underlying the tables and figures will be deposited in a permanent OSF archive before journal submission. Raw MAFW videos are third-party materials and are not redistributed; access is governed by the dataset provider. The deposit will include the derived summaries required to reproduce the empirical calibration analyses.

Code availability

Code for generating LOMM datasets, fitting the methods, reproducing the statistical summaries, and regenerating the tables and figures will accompany the permanent OSF deposit. The deposit will also provide the manuscript source and instructions for reproducing the analyses.

Authors' contributions

Aleksandar Tomašević contributed conceptualization, formal analysis, methodology, resources, software, validation, visualization, writing the original draft, and reviewing and editing the manuscript. Hudson Golino contributed formal analysis, methodology, resources, software, validation, visualization, writing the original draft, and reviewing and editing the manuscript. Alexander P. Christensen contributed methodology, software, writing the original draft, and reviewing and editing the manuscript.

Open Practices Statement

The study was not preregistered. Before journal submission, the authors will deposit the simulation and analysis code, manuscript source, data underlying the figures and tables, and derived empirical score summaries used in the calibration analysis in a permanent OSF archive. Raw MAFW videos are third-party materials governed by the dataset provider and will not be included.

References

Allison, B., & McKeown, G. (2023). Are facial expression technologies tools for emotion recognition or social interaction analysis, or both? *2023 11th International Conference on Affective Computing and Intelligent Interaction Workshops and Demos (ACIIW)*, 1–8. https://doi.org/10.1109/ACIIW59127.2023.10388163

Arriaga, O., Valdenegro-Toro, M., & Plöger, P. (2017). *Real-time Convolutional Neural Networks for Emotion and Gender Classification* (arXiv:1710.07557). arXiv. https://doi.org/10.48550/arXiv.1710.07557

Blondel, V. D., Guillaume, J.-L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, *2008*(10), P10008. https://doi.org/10.1088/1742-5468/2008/10/P10008

Boker, S. M. (2015). Adaptive equilibrium regulation: A balancing act in two timescales. *Journal for Person-Oriented Research*, *1*(1-2), 99–109. https://doi.org/10.17505/jpor.2015.10

Boker, S. M., Deboek, P. R., Edler, C., & Keel, P. (2010). Generalized local linear approximation of derivatives from time series. In S. M. Chow, E. Ferrer, & F. Hsieh (Eds.), *The notre dame series on quantitative methodology. Statistical methods for modeling human dynamics: An interdisciplinary dialogue* (pp. 161–178). Routledge/Taylor & Francis Group.

Boker, S. M., & Nesselroade, J. R. (2002). A Method for Modeling the Intrinsic Dynamics of Intraindividual Variability: Recovering the Parameters of Simulated Oscillators in Multi-Wave Panel Data. *Multivariate Behavioral Research*, *37*(1), 127–160. https://doi.org/10.1207/S15327906MBR3701_06

Epskamp, S. (2024). *graphicalVAR: Graphical VAR for experience sampling data*. https://CRAN.R-project.org/package=graphicalVAR

Epskamp, S., Waldorp, L. J., Mõttus, R., & Borsboom, D. (2018). The gaussian graphical model in cross-sectional and time-series data. *Multivariate Behavioral Research*, *53*(4), 453–480. https://doi.org/10.1080/00273171.2018.1454823

Ezerceli, Ö., & Eskil, M. T. (2022). Convolutional Neural Network (CNN) Algorithm Based Facial Emotion Recognition (FER) System for FER-2013 Dataset. *2022 International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME)*, 1–6. https://doi.org/10.1109/ICECCME55909.2022.9988371

Foteinopoulou, N. M., & Patras, I. (2023). *EmoCLIP: A Vision-Language Method for Zero-Shot Video Facial Expression Recognition* (arXiv:2310.16640). arXiv. https://doi.org/10.48550/arXiv.2310.16640

Gates, K. M., & Molenaar, P. C. M. (2012). Group search algorithm recovers effective connectivity maps for individuals in homogeneous and heterogeneous samples. *NeuroImage*, *63*(1), 310–319. https://doi.org/10.1016/j.neuroimage.2012.06.026

Golino, H. F., & Epskamp, S. (2017). Exploratory graph analysis: A new approach for estimating the number of dimensions in psychological research. *PloS One*, *12*(6), e0174035. https://doi.org/10.1371/journal.pone.0174035

Golino, H., Christensen, A. P., Moulder, R., Kim, S., & Boker, S. M. (2022). Modeling latent topics in social media using dynamic exploratory graph analysis: The case of the right-wing and left-wing trolls in the 2016 US elections. *Psychometrika*, *87*(1), 156–187.

Golino, H., Moulder, R., Shi, D., Christensen, A. P., Garrido, L. E., Nieto, M. D., Nesselroade, J., Sadana, R., Thiyagarajan, J. A., & Boker, S. M. (2021). Entropy fit indices: New fit measures for assessing the structure and dimensionality of multiple latent variables. *Multivariate Behavioral Research*, *56*(6), 874–902. https://doi.org/10.1080/00273171.2020.1779642

Golino, H., Shi, D., Garrido, L. E., Christensen, A. P., Nieto, M. D., Sadana, R., Thiyagarajan, J. A., & Martinez-Molina, A. (2020). Investigating the performance of exploratory graph analysis and traditional techniques to identify the number of latent factors: A simulation and tutorial. *Psychological Methods*, *25*(3), 292–320. https://doi.org/10.1037/met0000255

Goodfellow, I. J., Erhan, D., Carrier, P. L., Courville, A., Mirza, M., Hamner, B., Cukierski, W., Tang, Y., Thaler, D., Lee, D.-H., Zhou, Y., Ramaiah, C., Feng, F., Li, R., Wang, X., Athanasakis, D., Shawe-Taylor, J., Milakov, M., Park, J., … Bengio, Y. (2013). *Challenges in Representation Learning: A report on three machine learning contests* (arXiv:1307.0414). arXiv. https://doi.org/10.48550/arXiv.1307.0414

Halamová, J., Kanovský, M., Brockington, G., & Strnádelová, B. (2023). Automated facial expression

analysis of participants self-criticising via the two-chair technique: Exploring facial behavioral markers of self-criticism. *Frontiers in Psychology*, *14*. https://doi.org/10.3389/fpsyg.2023.1138916

Holiday, S., Hayes, J. L., Park, H., Lyu, Y., & Zhou, Y. (2023). A multimodal emotion perspective on social media influencer marketing: The effectiveness of influencer emotions, network size, and branding on consumer brand engagement using facial expression and linguistic analysis. *Journal of Interactive Marketing*, *58*(4), 414–439. https://doi.org/10.1177/10949968231171104

Hollenstein, T. (2007). State space grids: Analyzing dynamics across development. *International Journal of Behavioral Development*, *31*(4), 384–396. https://doi.org/10.1177/0165025407077765

Houben, M., Van Den Noortgate, W., & Kuppens, P. (2015). The relation between short-term emotion dynamics and psychological well-being: A meta-analysis. *Psychological Bulletin*, *141*(4), 901–930. https://doi.org/10.1037/a0038822

Kankaanpää, R., Ron, J. de, Hoekstra, R. H. A., & Bork, R. van. (2026). Comparing multiple-indicator approaches to account for measurement error in dynamic networks. *Cognitive Therapy and Research*. https://doi.org/10.1007/s10608-026-10719-0

Kollias, D., Tzirakis, P., Baird, A., Cowen, A., & Zafeiriou, S. (2023). ABAW: Valence-arousal estimation, expression recognition, action unit detection, and emotional reaction intensity estimation challenges. *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 5889–5898. https://doi.org/10.1109/CVPRW59228.2023.00626

Kuppens, P., & Verduyn, P. (2017). Emotion dynamics. *Current Opinion in Psychology*, *17*, 22–26. https://doi.org/10.1016/j.copsyc.2017.06.004

Lane, S. T., Gates, K. M., Pike, H. K., Beltz, A. M., & Wright, A. G. C. (2019). Uncovering general, shared, and unique temporal patterns in ambulatory assessment data. *Psychological Methods*, *24*(1), 54–69. https://doi.org/10.1037/met0000192

Li, H., Niu, H., Zhu, Z., & Zhao, F. (2023). *CLIPER: A Unified Vision-Language Framework for In-the-Wild Facial Expression Recognition* (arXiv:2303.00193). arXiv. https://doi.org/10.48550/arXiv.2303.00193

Li, Q., Liu, Z., Zhang, Z., Wang, Q., & Ma, M. (2024). Decoding group emotional dynamics in a web-based collaborative environment: A novel framework utilizing multi-person facial expression recognition. *International Journal of Human-Computer Interaction*, 1–19. https://doi.org/10.1080/10447318.2024.2338614

Li, S., & Deng, W. (2022). Deep Facial Expression Recognition: A Survey. *IEEE Transactions on Affective Computing*, *13*(3), 1195–1215. https://doi.org/10.1109/TAFFC.2020.2981446

Liu, Y., Dai, W., Feng, C., Wang, W., Yin, G., Zeng, J., & Shan, S. (2022). MAFW: A large-scale, multi-modal, compound affective database for dynamic facial expression recognition in the wild. *Proceedings of*

the 30th ACM International Conference on Multimedia, 24–32. https://doi.org/10.1145/3503161.3548190

Manelis-Baram, L., Barami, T., Ilan, M., Meiri, G., Menashe, I., Soskin, E., Sofer, C., & Dinstein, I. (2025). Comparing three algorithms of automated facial expression analysis in autistic children: Different sensitivities but consistent proportions. *Molecular Autism*, *16*(1). https://doi.org/10.1186/s13229-025-00685-x

Mehu, M., & Scherer, K. R. (2015). Emotion categories and dimensions in the facial communication of affect: An integrated approach. *Emotion*, *15*(6), 798–811. https://doi.org/10.1037/a0039416

Meng, Z., Liu, P., Cai, J., Han, S., & Tong, Y. (2017). Identity-Aware Convolutional Neural Network for Facial Expression Recognition. *2017 12th IEEE International Conference on Automatic Face & Gesture Recognition (FG 2017)*, 558–565. https://doi.org/10.1109/FG.2017.140

Mollahosseini, A., Hasani, B., & Mahoor, M. H. (2019). AffectNet: A database for facial expression, valence, and arousal computing in the wild. *IEEE Transactions on Affective Computing*, *10*(1), 18–31. https://doi.org/10.1109/TAFFC.2017.2740923

Nesselroade, J. R., McArdle, J. J., Aggen, S. H., & Meyers, J. M. (2002). Dynamic factor analysis models for representing process in multivariate time-series. In D. S. Moskowitz & S. L. Hershberger (Eds.), *Multivariate applications book series. Modeling intraindividual variability with repeated measures data: Methods and applications* (pp. 235--265). Lawrence Erlbaum Associates Publishers.

Ollero, M. J. F., Estrada, E., Hunter, M. D., & Cáncer, P. F. (2025). Characterizing affect dynamics with a damped linear oscillator model: Theoretical considerations and recommendations for individual-level applications. *Psychological Methods*, *30*(5), 1095–1112. https://doi.org/10.1037/met0000615

Plutchik, R. (2001). The nature of emotions. *American Scientist*, *89*(4), 344–350. https://doi.org/10.1511/2001.4.344

Pons, P., & Latapy, M. (2005). Computing communities in large networks using random walks. In Pi. Yolum, T. Güngör, F. Gürgen, & C. Özturan (Eds.), *Computer and information sciences - ISCIS 2005* (pp. 284–293). Springer Berlin Heidelberg. https://doi.org/10.1007/11569596_31

Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., & Sutskever, I. (2021). Learning transferable visual models from natural language supervision. *Proceedings of the 38th International Conference on Machine Learning*, 8748–8763.

Russell, J. A. (1980). A circumplex model of affect. *Journal of Personality and Social Psychology*, *39*(6), 1161–1178. https://doi.org/10.1037/h0077714

Russell, J. A. (2003). Core affect and the psychological construction of emotion. *Psychological Review*, *110*(1), 145–172. https://doi.org/10.1037/0033-295X.110.1.145

Russell, J. A., & Mehrabian, A. (1977). Evidence for a three-factor theory of emotions. *Journal of Research*

in Personality, *11*(3), 273–294. https://doi.org/10.1016/0092-6566(77)90037-X

Tomašević, A., Golino, H., & Christensen, A. (2026). transforEmotion: An open-source R package for emotion analysis using transformer-based generative AI models. *Computational Communication Research*, *8*(2), 1. https://doi.org/10.5117/CCR2026.2.2.TOMA

Traag, V. A., Waltman, L., & Eck, N. J. van. (2019). From louvain to leiden: Guaranteeing well-connected communities. *Scientific Reports*, *9*(1), 5233. https://doi.org/10.1038/s41598-019-41695-z

Wang, W., Sebe, N., & Lepri, B. (2022). *Rethinking the Learning Paradigm for Facial Expression Recognition* (arXiv:2209.15402). arXiv. https://arxiv.org/abs/2209.15402

Zhang, Z., Hamaker, E. L., & Nesselroade, J. R. (2008). Comparisons of four methods for estimating a dynamic factor model. *Structural Equation Modeling: A Multidisciplinary Journal*, *15*(3), 377–402. https://doi.org/doi.org/10.1080/10705510802154281

Appendix A: Complete Specification of LOMM

This appendix states LOMM in full. The main text gives the three equations a reader needs to follow the studies; everything omitted there appears here, in the same order as the corresponding main-text material. Nothing in this appendix changes any result. It is intended for readers who want to reimplement the generator or check the claims about what it can and cannot identify.

A.1 The Latent Transition

Equation 1 of the main text moves each dimension forward one observation interval. Its two components are

$$\Phi_f(\Delta) = \exp(\mathbf{A}_f\Delta), \qquad \mathbf{A}_f = \begin{bmatrix} 0 & 1 \\ \eta_f & \zeta_f \end{bmatrix} \qquad \text{and} \qquad \Gamma_f(\Delta) = \int_0^\Delta \exp(\mathbf{A}_f u)\mathbf{b}\, du, \quad \mathbf{b} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

For the nonzero position-feedback values used in the simulation grids, the input map has the equivalent exact form

$$\Gamma_f(\Delta) = \mathbf{A}_f^{-1}\{\Phi_f(\Delta) - \mathbf{I}_2\}\mathbf{b}.$$

The integral definition remains valid when $\mathbf{A}_f$ is singular, including $\eta_f = 0$. If a root-based closed form is used numerically, a separate repeated-root branch is required when $\zeta_f^2 + 4\eta_f$ is close to zero.

The state components $x_{if,n}$ and $v_{if,n}$ are the level and its rate of change at t_n; η_f is position feedback, ζ_f is velocity feedback, and σ_{qf} is the standard deviation of the acceleration input. On the interval $[t_n, t_{n+1})$ the underlying continuous-time model is $\ddot{x}_{if}(t) = \eta_f x_{if}(t) + \zeta_f \dot{x}_{if}(t) + q_{if,n}$ with $q_{if,n}$ held constant, so $\Phi_f(\Delta)$ propagates the deterministic state and $\Gamma_f(\Delta)$ loads the input. The inputs are independent across videos, dimensions, and transitions, and independent of initial states, activity schedules, and measurement errors. They are zero-order-hold acceleration inputs rather than increments of a continuous-time white-noise diffusion. The covariance added in one transition is $\Gamma_f(\Delta)\sigma_{qf}^2\Gamma_f(\Delta)^\top$.

A.2 Dynamic Regimes and Finite-Horizon Amplification

The characteristic roots for dimension f are

$$r_{f,\pm} = \frac{\zeta_f \pm \sqrt{\zeta_f^2 + 4\eta_f}}{2}.$$

Continuous-time asymptotic stability requires $\eta_f < 0$ and $\zeta_f < 0$, and the roots are oscillatory when $\zeta_f^2 + 4\eta_f < 0$. The exact discrete transition is asymptotically stable when

$$\rho\{\Phi_f(\Delta)\} = \exp\left[\Delta \max\{\mathrm{Re}(r_{f,+}), \mathrm{Re}(r_{f,-})\}\right] < 1.$$

Negative η_f supplies restoring acceleration and negative ζ_f attenuates velocity, whereas positive ζ_f produces amplification. How much amplification actually appears depends on the observed horizon. For T observations, $H = (T-1)\Delta$. Let $\alpha(\mathbf{A}_f) = \max\{\mathrm{Re}(r_{f,+}), \mathrm{Re}(r_{f,-})\}$. The quantity

$$g_f(H) = \exp\{H\alpha(\mathbf{A}_f)\}$$

is a modal exponential-rate factor over that horizon. It does not by itself measure nonnormal transient amplification or the polynomial factor that can occur at a defective repeated root. A norm-based finite-horizon state-amplification diagnostic is

$$G_f^{\max}(H) = \sup_{0 \le u \le H} \|\exp(\mathbf{A}_f u)\|_2,$$

and the corresponding level-only diagnostic is

$$G_{x,f}^{\max}(H) = \sup_{0 \le u \le H} \left\| \begin{bmatrix} 1 & 0 \end{bmatrix} \exp(\mathbf{A}_f u) \right\|_2.$$

The Euclidean matrix norm is coordinate-dependent. Because the state combines a level and a rate of change, whose components have different units, this diagnostic is conditional on the chosen state coordinates and should be interpreted only after fixing a nondimensionalization or an explicit weighted state norm. A coordinate-aware definition is

$$G_{f,M}^{\max}(H) = \sup_{0 \le u \le H} \left\| \mathbf{M}^{1/2} \exp(\mathbf{A}_f u) \mathbf{M}^{-1/2} \right\|_2,$$

for a stated positive-definite state metric $\mathbf{M}$; the unweighted $G_f^{\max}(H)$ corresponds to $\mathbf{M} = \mathbf{I}_2$. Reporting the roots and $g_f(H)$ characterizes the modal regime over the observed horizon, but $g_f(H)$ should not be read as total finite-horizon amplification. Reporting the activity schedule separately keeps amplification in the latent dynamics distinct from nonstationarity created by the activity layer.

A.3 Initial States

The system is uncoupled: η_f, ζ_f, and σ_{qf} may vary by dimension, and off-diagonal dynamic effects are omitted. Study 2's main benchmark initialized every dimension at $x_{if,1} = 0.25$ and $v_{if,1} = 2$. The sensitivity analysis used independent starting states. For a stable dimension, the stationary covariance $\mathbf{P}_f$ solves

$$\mathbf{P}_f = \Phi_f(\Delta)\mathbf{P}_f\Phi_f(\Delta)^\top + \Gamma_f(\Delta)\sigma_{qf}^2\Gamma_f(\Delta)^\top,$$

whose solution is unique when $\rho\{\Phi_f(\Delta)\} < 1$. Drawing $\mathbf{s}_{if,1} \sim \mathcal{N}(\mathbf{0}, \mathbf{P}_f)$ makes the state process stationary before the activity layer is applied; a time-varying activity envelope still renders $a_{if}(t)x_{if}(t)$ nonstationary. For a dimension with positive velocity feedback, the sensitivity draws $x_{if,1} \sim \text{Unif}(-0.25, 0.25)$ and $v_{if,1} \sim \text{Unif}(-2, 2)$. These independent starts remove the transient that all dimensions would otherwise share, while leaving the observed horizon unchanged.

A.4 The Activity Layer at a Transition

Away from a discontinuity in the activity value, the ordinary product rule applies,

$$\frac{d}{dt}\{a_{if}(t)x_{if}(t)\} = a'_{if}(t)x_{if}(t) + a_{if}(t)x'_{if}(t).$$

At an abrupt transition, the derivative of the activity value contains an impulsive component in the distributional sense. Define $D_n x = x_{n+1} - x_n$. The exact discrete identity is

$$D_n(a_{if}x_{if}) = a_{if,n+1}D_n x_{if} + (D_n a_{if})x_{if,n},$$

where the first term uses the post-transition activity value and the second uses the pre-transition state. The corresponding finite-difference identity is

$$\frac{\tilde{x}_{if,n+1} - \tilde{x}_{if,n}}{\Delta} = a_{if,n+1}\frac{x_{if,n+1} - x_{if,n}}{\Delta} + \frac{a_{if,n+1} - a_{if,n}}{\Delta}x_{if,n}.$$

The second term is what a derivative-based method sees at a switch even when the oscillator itself has not accelerated, which is the reason for the gradual-transition sensitivity analysis. This convention attaches the new activity value to the post-transition observation, so the activity layer multiplies each completed latent state by the schedule value at the same observation index.

A.5 The Observation Layer

Let J denote the number of positive score channels, let $d = J - 1$, and let $\mathbf{V} \in \mathbb{R}^{J \times d}$ be a fixed orthonormal isometric log-ratio (ILR) basis satisfying $\mathbf{V}^\top \mathbf{V} = \mathbf{I}_d$ and $\mathbf{V}^\top \mathbf{1}_J = \mathbf{0}$. The basis used here is a normalized Helmert basis constructed in the stated 11-channel order. In Equation 2 of the main text, $\mu \in \mathbb{R}^d$ is a fixed center, $\mathbf{b}_i \in \mathbb{R}^d$ is the clip baseline, $s_i > 0$ is the clip scale, $\kappa > 0$ is a global scale, and $\Lambda \in \mathbb{R}^{d \times m}$ maps the m activity-scaled latent positions into these coordinates. Conditional on the clip-specific terms and Σ_i, measurement errors are independent over time and across videos, and independent of latent states, initial states, activity schedules, and acceleration inputs, with $\varepsilon_{in} \mid \Sigma_i \sim \mathcal{N}(\mathbf{0}, \Sigma_i)$. Dependence across coordinates follows Σ_i. The clip scale multiplies both signal and residual, whereas the baseline is additive.

The three baseline constructions differ only in where $\mathbf{b}_i$ is allowed to live. Study 1 draws it from a zero-mean Gaussian whose covariance is estimated from training clips in the full coordinate space. Study 2's between-clip-variability analysis instead uses

$$\mathbf{b}_i = \Lambda \beta_i, \qquad E(\beta_i) = \mathbf{0},$$

so that between-clip variation stays within the space spanned by the loadings and does not introduce structure outside the factor model. Study 2's main conditions fix $\mathbf{b}_i = \mathbf{0}$ and $s_i = 1$, giving the exactly nested homogeneous case

$$\mathbf{u}_{in} = \mu + \kappa \left(\Lambda \tilde{\mathbf{x}}_{in} + \varepsilon_{in} \right).$$

Study 2's continuous indicators are the pre-link quantity

$$\mathbf{z}_{in} = \Lambda_i \tilde{\mathbf{x}}_{in} + \varepsilon_{in},$$

in which each of the $p = mk$ indicators loads on one dimension and has diagonal measurement error.

For bounded output, LOMM maps the coordinates to the open probability simplex. The two log-ratio transformations are

$$\mathrm{clr}(\mathbf{p}) = \log \mathbf{p} - J^{-1} \{ \mathbf{1}_J^\top \log(\mathbf{p}) \} \mathbf{1}_J, \qquad \mathrm{ilr}_{\mathbf{V}}(\mathbf{p}) = \mathbf{V}^\top \mathrm{clr}(\mathbf{p}),$$

with the logarithm applied componentwise. Applying Equation 3 of the main text then gives

$$\mathrm{ilr}_{\mathbf{V}}(\mathbf{p}_{in}) = \frac{\mathbf{u}_{in}}{\tau}, \qquad \mathrm{clr}(\mathbf{p}_{in}) = \frac{\mathbf{V}\mathbf{u}_{in}}{\tau},$$

the second because $\mathbf{V}\mathbf{u}_{in}$ lies in the zero-sum log-ratio subspace. The fixed basis excludes the unidentified common-logit direction and the softmax produces strictly positive rows summing to one, so the scores identify the scaled coordinates $\mathbf{u}_{in}/\tau$ exactly. A common-logit component remains unobservable, and the unscaled coordinates and τ are not separately identified without an external scale constraint.

A.6 Converting Period and Damping to Feedback Parameters

Study 1's search was parameterized by natural period and damping ratio rather than by η_f and ζ_f directly. If P_f is the natural period in observation intervals and ξ_f is the damping ratio, then

$$\omega_{n,f} = \frac{2\pi}{P_f \Delta}, \qquad \eta_f = -\omega_{n,f}^2, \qquad \zeta_f = -2\xi_f \omega_{n,f},$$

and the damped angular frequency is $\omega_{d,f} = \omega_{n,f}\sqrt{1-\xi_f^2}$. With $P_f = 12$, $\xi_f = 0.5$, and $\Delta = 1$, these give $\eta_f = -0.2742$ and $\zeta_f = -0.5236$; scaling the inputs to unit stationary position variance gives $\sigma_{qf} = 0.5418$.

A.7 Identifiability

For any nonsingular diagonal matrix $\mathbf{D}$,

$$\Lambda\tilde{\mathbf{x}}_{in} = (\Lambda\mathbf{D}^{-1})(\mathbf{D}\tilde{\mathbf{x}}_{in}),$$

so loading magnitude and latent-trajectory scale are not jointly identified without an explicit scale constraint. Activity gain and acceleration-input scale contribute to latent amplitude in the same way. The softmax layer adds invariance to a common location shift and a tradeoff between the temperature and the coordinate scale. Dimension labels are recoverable only up to permutation.

Sampling cadence adds an aliasing restriction. For an underdamped dimension, write

$$r_{f,\pm} = \frac{\zeta_f}{2} \pm i\omega_f, \qquad \omega_f = \frac{1}{2}\sqrt{-(\zeta_f^2 + 4\eta_f)}.$$

At observation times $t_n = (n-1)\Delta$, the frequencies ω_f and $\omega_f + 2\pi q/\Delta$ for integer q generate the same sampled sine and cosine sequences after an appropriate change in initial velocity. Identifying continuous-time oscillator parameters from data therefore requires a cadence-dependent restriction such as

$$0 < \omega_f < \frac{\pi}{\Delta},$$

together with a specified acceleration-input model and constraints on the latent state basis. Estimating LOMM from data would in total require sign and scale constraints on the dimensions, normalization of the activity schedule, anchoring of either the loading scale or the input scale, a reference-logit or sum-to-zero constraint, a fixed or otherwise identified temperature, and this frequency restriction. Every oscillatory mode used here satisfies the restriction: Study 1 uses $\Delta = 1$, giving the most restrictive bound $\pi \approx 3.14$ against a largest angular frequency of approximately 0.78, and Study 2 uses $\Delta \in \{0.25, 0.4\}$, giving bounds of approximately 12.57 and 7.85 against a largest angular frequency of approximately 1.04. The simulations do not estimate any of these parameters.

Appendix B: What Each Method's Network Looks Like in the Population

This appendix works out what the network each method estimates would look like in the population, for Study 2's continuous indicators with every video clip sharing one measurement model. Its purpose is to explain why derivative-based and lagged-residual networks turn out to be more similar than their descriptions suggest. Throughout, a *block* is the set of k indicators loading on a single dimension, so recovering the blocks is the same thing as recovering the assignment of variables to dimensions.

Each method proceeds in four stages: it estimates a structure, converts that structure to a network, applies community detection, and may fail to produce a usable result at all. Estimation and conversion error are treated together, conditional on a usable fit; community detection acts on the network afterward and is not part of the network's error; a failed fit has no network to perturb.

Each analysis can be written as

$$\hat{\mathbf{S}}^{(r)} \xrightarrow{\Pi_r} \widetilde{\mathbf{W}}^{(r)} \xrightarrow{\mathcal{C}_r} \widehat{\mathcal{B}}^{(r)},$$

where $\hat{\mathbf{S}}^{(r)}$ is the fitted method-specific structure and Π_r converts it to the graph $\widetilde{\mathbf{W}}^{(r)}$. The community-detection algorithm $\mathcal{C}_r$ returns the partition $\widehat{\mathcal{B}}^{(r)}$. Let U_r denote the event that a usable graph exists. Conditional on $U_r = 1$,

$$\widetilde{\mathbf{W}}^{(r)} = \mathbf{W}_*^{(r)} + \Xi^{(r)},$$

where $\mathbf{W}_*^{(r)}$ is the population structural matrix induced by the method's graph construction. The perturbation $\Xi^{(r)}$ includes finite-sample variation, derivative or model estimation, regularization, graph extraction, dependence-induced variance inflation, and systematic deviation from the symmetric target. The community algorithm acts after the graph is formed. When $U_r = 0$, there is no estimated graph to perturb.

The derivation has three implications. First, in the ideal case both the DynEGA and the GraphicalVAR population networks preserve the true blocks, provided the relevant cross-block moments are zero. Second, their positive within-block partial-correlation edges share the same ceiling of $1/(k-1)$. How closely a given edge approaches that ceiling depends on the signal-to-noise ratio each method achieves, and the algebra alone does not say which method achieves more. Third, GIMME stands on different footing, because what it

Table B1: Assumptions behind the Appendix B population results and the corresponding sensitivity analyses.

Result or layer	Main assumptions	Sensitivities and interpretation
Exact block support	Zero pooled cross-block moments after filtering or projection; one primary loading per indicator; independent measurement errors	Cross-loadings and correlated acceleration inputs introduce cross-block dependence; within-block residual dependence violates local error independence and can reinforce or distort the block signal
Per-channel filter invariance	The same linear temporal filter is applied to every indicator, with fixed boundary handling	Short records, embedding selection, and gate transitions can change finite-sample filter behavior
Within-block edge formula	Exchangeable loadings and error variances within a block, with positive filtered signal variance	Loading and variance jitter perturb edge equality while preserving the ideal support target
Connected-component recovery	The weakest population within-block edge exceeds total graph perturbation conditional on a usable fit	Regularization, sampling error, and estimation contribute to perturbation; fit failure removes the graph
GIMME recovery decomposition	A usable contemporaneous graph can be extracted and partitioned	Convergence, timeout, extraction, symmetrization, and completion are separate analysis stages

recovers also depends on path search, convergence, extraction, averaging, network conversion, and whether it completes at all. Two things this derivation does not cover: Study 1's bounded scores and their log-ratio coordinates, and the between-clip-variability analysis, which adds mean differences across clips that fall outside the zero-mean target assumed here.

B.1 Linear Block Target and Pooled Moments

In this appendix, Y_j corresponds to a main-text pre-link score channel z_j, X_f to the ungated latent level x_f, and $Z_f = a_f X_f$ to the activity-scaled latent level $\tilde{x}_f$. Video indices and loading and variance jitter are omitted to define a symmetric population target.

Consider the population target for the main continuous-indicator benchmark, with m latent dimensions, k observed indicators per dimension, and $p = mk$ observed score channels. Let $B_1, \ldots, B_m$ denote the true indicator blocks. For indicator $j \in B_f$, use the equal-loading, equal-error measurement model

$$Y_{j,n} = \lambda Z_{f,n} + \varepsilon_{j,n}, \qquad Z_{f,n} = a_{f,n} X_{f,n},$$

where $Z_{f,n}$ is the activity-scaled latent oscillator for dimension f, $\lambda > 0$, and $\varepsilon_{j,n}$ is measurement error with variance $\psi = s_\varepsilon^2$. Measurement errors are independent across indicators and observations in this target, so $\mathrm{Cov}(\varepsilon_{j,n}, \varepsilon_{\ell,h}) = 0$ whenever $j \neq \ell$ or $n \neq h$.

Let $\mathcal{T}$ denote the rows of the common simulated dataset. Method r retains a subset $\mathcal{R}_r \subseteq \mathcal{T}$. The GLLA derivative series loses rows according to the selected embedding length and boundary rule, whereas the lag-1 projection retains within-video lag pairs. The simulated dataset and physical horizon are common, but the retained rows differ between methods; no common row intersection is imposed. Population moments for method r are taken over $\mathcal{R}_r$. For any quantity g,

$$\langle g \rangle_{\mathcal{R}_r} = \frac{1}{|\mathcal{R}_r|} \sum_{(i,n)\in\mathcal{R}_r} \mathbb{E}(g_{in}),$$

and, for any filtered process U,

$$\mathrm{Var}_{\mathcal{R}_r}(U) = \left\langle \{U - \langle U \rangle_{\mathcal{R}_r}\}^2 \right\rangle_{\mathcal{R}_r}.$$

All variances and covariances in this appendix are time-averaged centered second moments over the rows retained by the method in question. We write this set as $\mathcal{R}$ when the dependence on the method is suppressed. Because the gated and finite-horizon-amplifying processes are nonstationary, different retained windows can yield different $V_{L,f}$, $V_{+,f}$, $V_{-,f}$, C_f, and $\theta_{r,f}$ on the same simulated dataset. The choice of retained rows therefore changes the population moments.

Exact block support is imposed through zero pooled cross-block moments over the retained rows. For every pair $f \neq g$ and every method-relevant filter or projection, the corresponding centered cross-block covariance is assumed to be zero. One sufficient set of conditions is that oscillator states are independent across factors at all relevant times and have zero conditional mean at every retained row. Activity schedules must also be deterministic or independent of the states and shocks, and dynamic shocks must be independent across factors. Uncoupled dynamic equations alone do not imply zero pooled cross-block covariance.

Coordinated activity schedules do not create cross-block covariance when factor states have zero conditional means. A shared deterministic initialization, however, can create common mean transients and violate the pooled zero-cross-covariance condition even when the dynamic equations and shocks are uncoupled. The common-start benchmark is therefore a structured departure from the ideal target. The independent zero-mean initialization sensitivity removes this channel.

Activity gating makes Z_f nonstationary by design, which is why the covariances in this appendix are defined as pooled centered moments rather than as stationary second moments. Loading jitter, item-level variance jitter, switching schedules, initialization transients, finite-sample activity imbalance, and estimator behavior

are treated as departures or perturbations around this symmetric target.

B.2 Filtered Series and Two Network Mappings

The derivation uses two population network mappings for a per-channel filtered series. DynEGA estimates local derivatives of every channel; a lag difference is a simple analytic example of the same channelwise operation. GraphicalVAR is treated separately in B.3 because its lag-1 residuals come from a multivariate projection rather than an identical filter applied to each channel.

Let L denote a fixed per-channel linear filter with coefficients $\ell_0, \ldots, \ell_q$, identical boundary handling for every channel, and no window crossing a video boundary. Applying L to the measurement model for $j \in B_f$ gives

$$(LY_j)_n = \lambda\,(LZ_f)_n + (L\varepsilon_j)_n,$$

so the filtered series keeps the one-factor block structure. Write

$$V_{L,f} = \mathrm{Var}_{\mathcal{R}}(LZ_f), \qquad \psi_L = \mathrm{Var}_{\mathcal{R}}(L\varepsilon_j)$$

for the filtered common-signal variance and the filtered noise variance, both centered over the rows retained by the corresponding method. For independent and identically distributed (iid) base measurement error,

$$\psi_L = \psi \sum_{r=0}^{q} \ell_r^2,$$

and overlapping windows induce the lag covariance

$$\mathrm{Cov}\{(L\varepsilon_j)_n, (L\varepsilon_j)_{n+h}\} = \psi \sum_{r} \ell_r \ell_{r-h}.$$

Filtering preserves cross-channel error independence under the ideal error model, so the contemporaneous same-row block covariance remains diagonal plus rank one. Overlapping windows generally make filtered errors serially correlated. This serial dependence does not change the ideal same-row block covariance. It can, however, change the effective sample size, regularization behavior, and sampling variability. Section B.4 therefore treats serial dependence as a source of estimation error rather than incorporating it into the same-row population matrix.

Write $\mathbf{J}_k = \mathbf{1}_k\mathbf{1}_k^\top$. The within-block covariance matrix of the filtered channels is

$$\Sigma_{L,f} = \lambda^2 V_{L,f}\,\mathbf{J}_k + \psi_L \mathbf{I}_k, \qquad \theta_{L,f} = \frac{\lambda^2 V_{L,f}}{\psi_L},$$

provided the cross-block filtered moments vanish. Define

$$C_{L,fg} = \left\langle \{LZ_f - \mu_{L,f}\}\{LZ_g - \mu_{L,g}\} \right\rangle_{\mathcal{R}}, \qquad \mu_{L,f} = \langle LZ_f \rangle_{\mathcal{R}}.$$

The block-diagonal covariance and precision formulas in this section require $C_{L,fg} = 0$ for all $f \neq g$: exact block support is conditional on these zero pooled cross-block moments after filtering. The filtered signal-to-noise ratio $\theta_{L,f}$ is the only quantity through which the filter enters the population graph.

The filtered series admits two network mappings. The marginal-correlation graph uses correlations directly. For two distinct indicators $j, \ell \in B_f$,

$$w_f^M(L) = \left|\mathrm{Cor}(LY_j, LY_\ell)\right| = \frac{\theta_{L,f}}{1 + \theta_{L,f}}.$$

The partial-correlation graph conditions each pair on all other retained observed channels. Under exact block diagonality, the other blocks do not change the within-block calculation. Using the Sherman–Morrison identity,

$$\Sigma_{L,f}^{-1} = \psi_L^{-1}\left[\mathbf{I}_k - \frac{\theta_{L,f}}{1 + k\theta_{L,f}}\mathbf{J}_k\right],$$

and the absolute partial correlation between two indicators in the same block is

$$w_f^P(L) = \frac{\theta_{L,f}}{1 + (k-1)\theta_{L,f}}.$$

Because activity gating makes the row distribution time-varying, the inverse of the pooled covariance defines an algebraic partial-correlation graph. It is not interpreted as the conditional-independence graph of one stationary Gaussian process. Under the zero cross-block moment condition, the filtered precision matrix is block-diagonal and cross-block partial correlations are exactly zero. Both graph mappings therefore preserve the generating block support for the per-channel filtered population matrix. For any per-channel filter, the resulting graph has block-clique form,

$$\mathbf{W}_* = \text{blockdiag}\left(w_1(\mathbf{J}_k - \mathbf{I}_k), \ldots, w_m(\mathbf{J}_k - \mathbf{I}_k)\right), \qquad w_f \in \{w_f^M(L),\, w_f^P(L)\}.$$

The two mappings retain different amounts of within-block signal. For the same signal-to-noise ratio θ,

$$\frac{w^P(\theta)}{w^M(\theta)} = \frac{1+\theta}{1+(k-1)\theta},$$

so $w^P(\theta) < w^M(\theta)$ whenever $k > 2$ and $\theta > 0$. As $\theta \to \infty$,

$$w^P(\theta) \to \frac{1}{k-1}, \qquad w^M(\theta) \to 1.$$

The partial-correlation ceiling is a general consequence of this exchangeable block geometry:

$$\sup_{\theta > 0} w^P(\theta) = \frac{1}{k-1}.$$

Thus, even an arbitrarily strong common-source signal cannot produce a larger within-block partial-correlation edge under the equal-loading, equal-error target. With four indicators per dimension, the ceiling is 1/3. Both mappings increase with θ. The temporal filter sets $\theta_{L,f}$, and the graph mapping converts that ratio to an edge weight.

Figure B1 compares the marginal- and partial-correlation mappings for the benchmark's four indicators per dimension. The marginal mapping is shown only as a reference before conditioning; neither DynEGA nor GraphicalVAR uses it. The shaded band marks the raw loading-to-error-variance ratio $\lambda^2/s_\varepsilon^2$ in the benchmark rather than a calibrated range of $\theta_{L,f}$. With independent measurement error and a fixed filter, $\theta_{L,f} = (\lambda^2/\psi)V_{L,f}/\sum_r \ell_r^2$. Its position on the θ axis also depends on $V_{L,f}/\sum_r \ell_r^2$, which varies across scenarios and filters. Strong measurement conditions widen the common-source signal margin, but the partial-correlation edge remains bounded by 1/3 in the four-indicator target.

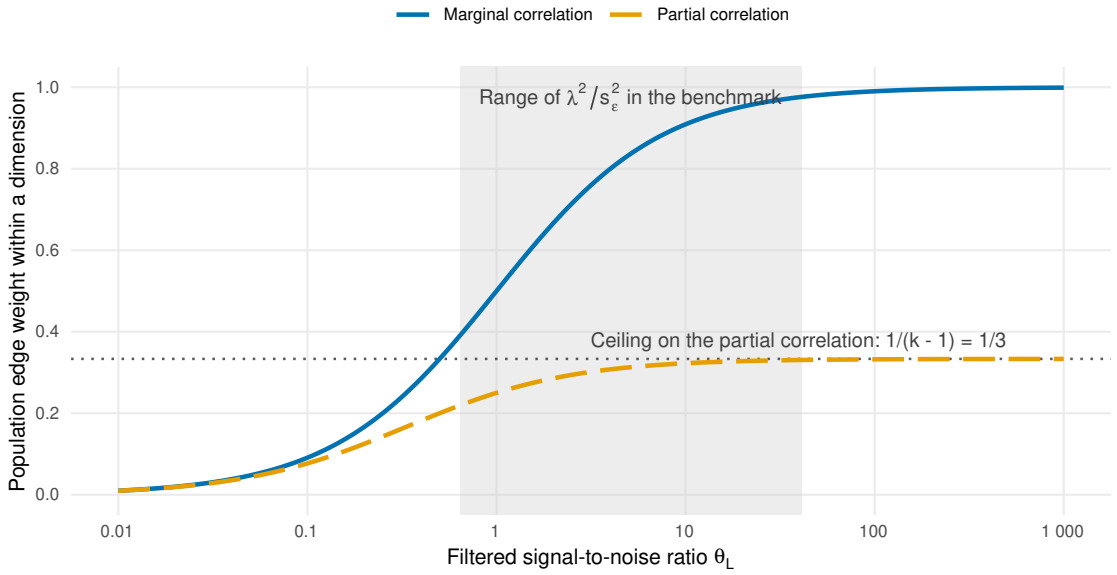


Figure B1: Population within-block edge weight as a function of the filtered signal-to-noise ratio for the marginal- and partial-correlation mappings, with four indicators per dimension. The dotted line marks the partial-correlation ceiling $1/(k-1) = 1/3$ under the equal-loading, equal-error target. The shaded band shows the raw loading-to-error-variance ratios $\lambda^2/s_\varepsilon^2$ in the matched simulation grid for reference. It is not a calibrated range of $\theta_{L,f}$, which also depends on the filtered common-signal variance $V_{L,f}$ and the filter noise gain $\sum_r \ell_r^2$.

B.3 Method-Specific Filters and Signal-to-Noise Ratios

The compared methods combine temporal transformations with graph construction. DynEGA estimates generalized local linear approximation (GLLA) derivatives of each score channel over a time-delay embedding window. It then estimates an EBICglasso regularized partial-correlation network from the derivative series. The analysis uses one observation interval for the GLLA spacing parameter; physical time is represented separately by $\Delta = 0.4$. With first-order derivatives, a fixed embedding, fixed retained rows, and no zero-variance jitter, conversion to derivatives per unit of physical time applies the same positive scale factor to every channel. The derivative correlation matrix is therefore unchanged. Zero-variance jitter would break this exact invariance if the added noise were not scaled with the derivatives.

B.3.1 DynEGA and a fixed derivative filter

For a fixed selected GLLA derivative filter with coefficient vector ℓ, let $\mathbf{R}_{Z,f}$ denote the covariance matrix of the latent values entering the embedding window, centered over the rows retained by the derivative filter. Under iid white measurement error, the population filtered signal-to-noise ratio is

$$\theta_{\mathrm{GLLA},f} = \frac{\lambda^2 \ell^\top \mathbf{R}_{Z,f} \ell}{\psi\, \ell^\top \ell}.$$

This population expression is conditional on the chosen embedding and boundary rule. Embedding selection and regularized estimation belong to the finite-sample analysis. DynEGA's population graph is obtained by applying the partial-correlation mapping to this ratio, $w_f^P(\theta_{\mathrm{GLLA},f})$.

A lag-h difference, $D_h Y_{j,n} = Y_{j,n+h} - Y_{j,n}$, gives a transparent analytic stand-in for that filter, with

$$\theta_f^{\Delta_h} = \frac{\lambda^2 \operatorname{Var}_{\mathcal{K}}\{Z_{f,n+h} - Z_{f,n}\}}{2\psi},$$

because independent measurement errors contribute 2ψ to the variance of every differenced channel. Both the lag difference and GLLA derivative estimation operate channel by channel. GLLA can reduce white measurement-noise variance by averaging over an embedding window, but it can also attenuate latent change components. Its signal-to-noise ratio is therefore scenario and embedding dependent.

B.3.2 GraphicalVAR lag-1 projection with finite-clip centering

GraphicalVAR reads partial correlations from the residual covariance of a lag-1 vector autoregression. Its population object is defined here by an unpenalized lag-1 linear projection, $\mathcal{P}_{\mathrm{VAR}}$, rather than a per-channel filter. The fitted analysis first standardizes each observed channel across the pooled data and then centers it within each clip before constructing lag pairs. The parameters below are therefore on the globally standardized scale immediately before within-clip centering. Centering changes both the contemporaneous error variance and the lag-1 error covariance.

For a clip with T observations, let

$$\tilde{\varepsilon}_{j,n} = \varepsilon_{j,n} - \bar{\varepsilon}_j, \qquad \bar{\varepsilon}_j = \frac{1}{T}\sum_{h=1}^{T} \varepsilon_{j,h}.$$

If the uncentered errors are independent with variance ψ, then

$$\operatorname{Var}(\tilde{\varepsilon}_{j,n}) = \psi\left(1 - \frac{1}{T}\right), \qquad \operatorname{Cov}(\tilde{\varepsilon}_{j,n}, \tilde{\varepsilon}_{j,n-1}) = -\frac{\psi}{T}.$$

No lag pair crosses a clip boundary. Let $\widetilde{Z}_{f,n} = Z_{f,n} - \bar{Z}_{f,i}$ denote the within-clip-centered latent signal for the clip containing observation n. For one symmetric block, define

$$V_{-,f} = \mathbb{E}_{\mathcal{R}}(\widetilde{Z}^2_{f,n-1}), \qquad V_{+,f} = \mathbb{E}_{\mathcal{R}}(\widetilde{Z}^2_{f,n}), \qquad C_f = \mathbb{E}_{\mathcal{R}}(\widetilde{Z}_{f,n}\widetilde{Z}_{f,n-1}).$$

The block moment matrices have the compound-symmetric form

$$\mathbf{S}_{--,f} = a_{-,f}\mathbf{J}_k + b_-\mathbf{I}_k, \qquad \mathbf{S}_{+-,f} = c_f\mathbf{J}_k + d\mathbf{I}_k, \qquad \mathbf{S}_{++,f} = a_{+,f}\mathbf{J}_k + b_+\mathbf{I}_k,$$

where

$$a_{-,f} = \lambda^2 V_{-,f}, \qquad a_{+,f} = \lambda^2 V_{+,f}, \qquad c_f = \lambda^2 C_f.$$

For equal-length clips and iid errors before centering,

$$b_- = b_+ = \psi\left(1 - \frac{1}{T}\right), \qquad d = -\frac{\psi}{T}.$$

The three moment matrices share two eigenspaces: the common direction spanned by $\mathbf{1}_k$ and its $(k-1)$-dimensional orthogonal complement. The lag-1 projection residual covariance

$$\Omega_{\mathrm{VAR},f} = \mathbf{S}_{++,f} - \mathbf{S}_{+-,f}\mathbf{S}^{-1}_{--,f}\mathbf{S}_{-+,f}$$

therefore has orthogonal and common-direction eigenvalues

$$r_{\perp,f} = b_+ - \frac{d^2}{b_-}, \qquad r_{\parallel,f} = b_+ + ka_{+,f} - \frac{(d + kc_f)^2}{b_- + ka_{-,f}}.$$

Equivalently,

$$\Omega_{\mathrm{VAR},f} = r_{\perp,f}\mathbf{I}_k + \frac{r_{\parallel,f} - r_{\perp,f}}{k}\mathbf{J}_k.$$

When $r_{\parallel,f} > r_{\perp,f} > 0$, define

$$\theta^{\mathrm{centered}}_{\mathrm{VAR},f} = \frac{r_{\parallel,f} - r_{\perp,f}}{kr_{\perp,f}}.$$

The positive within-block partial-correlation edge then follows the same compression map as B.2:

$$w^{P}_{\mathrm{VAR},f} = \frac{\theta^{\mathrm{centered}}_{\mathrm{VAR},f}}{1 + (k-1)\theta^{\mathrm{centered}}_{\mathrm{VAR},f}}.$$

Finite-clip centering preserves the diagonal-plus-rank-one residual form under the symmetric target, but it changes the signal-to-noise parameter in the partial-correlation mapping. If $r_{\parallel,f} \leq r_{\perp,f}$, the positive-edge graph does not contain the positive within-block signal assumed by the ceiling argument. The residual covariance should be called an innovation covariance only if the lag-1 VAR supplies the relevant innovation process. Oscillator position need not follow a VAR(1), so its projection residuals may remain serially correlated. The result also assumes zero cross-block lag-0 and lag-1 moments. It does not extend directly to cross-loadings, correlated factors, correlated errors, heterogeneous serial measurement error, or lag construction across clip boundaries.

B.3.3 Uncentered stationary special case and claim boundary

Setting $d = 0$ and $b_- = b_+ = \psi$ removes within-clip centering and recovers the simpler residual form

$$\Omega_{\mathrm{VAR},f} = \psi \mathbf{I}_k + c_{\mathrm{VAR},f}\mathbf{J}_k, \qquad c_{\mathrm{VAR},f} = \lambda^2 V_{+,f} - \frac{k\lambda^4 C_f^2}{\psi + k\lambda^2 V_{-,f}}.$$

By Cauchy–Schwarz, $C_f^2 \leq V_{+,f}V_{-,f}$, and therefore

$$c_{\mathrm{VAR},f} \geq \frac{\lambda^2 V_{+,f}\psi}{\psi + k\lambda^2 V_{-,f}} > 0.$$

The associated ratio is $\theta_{\mathrm{VAR},f} = c_{\mathrm{VAR},f}/\psi$. This uncentered form is a theoretical special case rather than the preprocessing used in the fitted GraphicalVAR analysis. If the latent process is also stationary, with

$$V_{+,f} = V_{-,f} = V_f, \qquad C_f = \rho_{1,f}V_f, \qquad \theta_f = \frac{\lambda^2 V_f}{\psi},$$

the uncentered lag-1 residual ratio is

$$\theta_{\mathrm{VAR},f} = \theta_f \left[1 - \rho^2_{1,f}\frac{k\theta_f}{1 + k\theta_f}\right],$$

while a lag-1 difference gives

$$\theta_{\Delta_1,f} = \theta_f(1 - \rho_{1,f}).$$

For $0 < \rho_{1,f} < 1$, $\theta_{\mathrm{VAR},f} > \theta_{\Delta_1,f}$. A first difference therefore has lower signal-to-noise than the lag-1 residual in this special case. This result rules out an automatic population advantage for differencing. It does not rank a selected GLLA derivative against the fitted, within-clip-centered GraphicalVAR analysis. As $\rho_{1,f} \to 1$,

$$\theta_{\mathrm{VAR},f} \to \frac{\theta_f}{1 + k\theta_f},$$

and, if $\theta_f \to \infty$ in that limit,

$$w^P_{\mathrm{VAR},f} \to \frac{1}{2k - 1},$$

whereas for fixed $\rho_{1,f} < 1$ and $\theta_f \to \infty$ the within-block edge instead approaches the shared $1/(k-1)$ ceiling under the symmetric target.

The population comparison between DynEGA and GraphicalVAR is consequently limited. On the positive branch, both methods produce partial-correlation edges and share the $1/(k-1)$ ceiling under the equal-loading, equal-error target. Both also preserve block support when their relevant cross-block moments are zero. The block algebra does not order $\theta_{\mathrm{GLLA},f}$ and $\theta^{\mathrm{centered}}_{\mathrm{VAR},f}$. A larger GLLA ratio would require embedding-window noise averaging to outweigh attenuation of the latent derivative signal. That balance must be evaluated from oracle or realized filtered covariances. Section B.4 collects the remaining estimation and graph-conversion differences in a perturbation margin.

B.4 Shared Perturbation Margin

Let a method's population graph have block-clique form

$$\mathbf{W}^{(r)}_* = \mathrm{blockdiag}\left(w^{(r)}_1(\mathbf{J}_k - \mathbf{I}_k), \ldots, w^{(r)}_m(\mathbf{J}_k - \mathbf{I}_k)\right),$$

where edge weights are on the nonnegative scale supplied to community detection and all m blocks share the same size k. For DynEGA and GraphicalVAR, $w^{(r)}_f$ equals $w^P_f(\theta_{\mathrm{GLLA},f})$ or $w^P_f(\theta^{\mathrm{centered}}_{\mathrm{VAR},f})$ from B.3. Each quantity is evaluated on the rows retained by the corresponding method and restricted to the positive branch. Conditional on $U_r = 1$, define the perturbation after method-specific graph conversion as

$$\delta_r = \|\widetilde{\mathbf{W}}^{(r)} - \mathbf{W}_*^{(r)}\|_{\max} = \max_{j \neq \ell} \left|\widetilde{W}_{j\ell}^{(r)} - W_{*,j\ell}^{(r)}\right|.$$

For any node $j \in B_f$, the estimated total connection to its true block is at least

$$(k-1)(w_f^{(r)} - \delta_r),$$

whereas its estimated total connection to any wrong block is at most

$$k\delta_r.$$

Thus each node has stronger total connection to its true block than to any wrong block whenever

$$(k-1)(w_{\min}^{(r)} - \delta_r) > k\delta_r,$$

or equivalently

$$\delta_r < \frac{k-1}{2k-1} w_{\min}^{(r)}.$$

For the matched grid with $k = 4$, the sufficient node-level block-preference and threshold-separation condition is

$$\delta_r < \frac{3}{7} w_{\min}^{(r)},$$

conditional on a usable graph and the exact block-support population target. This margin does not guarantee that Louvain, Leiden, or Walktrap recover the blocks, because exact recovery depends on the implemented community-detection algorithm. The bound has a narrower consequence. Because $\frac{k-1}{2k-1} < \frac{1}{2}$, it places every estimated cross-block edge below every estimated within-block edge. Any threshold $\tau \in (\delta_r,\, w_{\min}^{(r)} - \delta_r)$ then deletes all cross-block edges and retains each block as a connected clique. The connected components of the thresholded graph reproduce the generating blocks exactly. Threshold separation alone needs only $\delta_r < w_{\min}^{(r)}/2$.

The margin treats perturbation as symmetric noise on every edge, which makes δ_r a worst-case envelope.

Regularized estimation acts asymmetrically: glasso regularization tends to shrink precision off-diagonal entries toward zero and can remove weak edges, although its effect on each normalized partial correlation need not be monotone. In practice it can reduce $w_{\min}^{(r)}$ while setting weak cross-block edges to zero, which can support block separation.

Under the equal-loading, equal-error symmetric population target, every positive partial-correlation mapping of the diagonal-plus-rank-one block form has the ceiling $1/(k-1)$. DynEGA and GraphicalVAR both have this form under their respective assumptions, so $w_{\min}^{(r)} \leq 1/(k-1)$ for both. With $k = 4$, the sufficient perturbation margin approaches at most

$$\delta_r < \frac{3}{7} \cdot \frac{1}{3} = \frac{1}{7}$$

in the high-signal limit of the symmetric target. The comparison therefore depends on two quantities: how closely each filtered or projected ratio brings $w_{\min}^{(r)}$ to the shared ceiling, and the realized perturbation δ_r. An observed recovery ordering does not identify which method has the larger population signal-to-noise ratio because estimation, regularization, graph conversion, and feasibility also differ between methods.

The margin also describes block fragility. Activity gating enters through $V_{L,f}$, the time-averaged centered variance of the filtered activity-scaled process over the retained rows. A dimension held at background gain therefore contributes a smaller $\theta_{L,f}$ and a weaker within-block edge, all else equal. The quantities $V_{L,f}$, $w_{\min}^{(r)}$, and $\mathbf{W}_*^{(r)}$ remain conditional on the rows retained by each method. This conditioning is especially important for finite-horizon-amplifying factors because their moments grow across the observation window.

In the single-dominance families, background gains are 0.05–0.08 and active gains are 0.35–1.25 (Appendix F). Under comparable ungated filtered variances and error levels, the background gain is expected to reduce $\theta_{L,f}$. The weakest block is the factor with the smallest scenario-specific $V_{L,f}$ and must be identified from the simulated dataset or a population calculation. In the low-expression family, every dimension remains near background activity. The algebra therefore predicts a smaller population separation margin when the activity schedule weakens the least-active block. The recovery patterns in Figure 6 are consistent with this prediction, but the simulation does not estimate δ_r and cannot identify this mechanism on its own. For GIMME, the activity envelope is one of several influences because the resulting graph also depends on path search, convergence, extraction, averaging, graph conversion, and completion.

B.5 GIMME Graph Conversion

The GIMME comparator has a different mathematical status. Let Ψ_i^G denote the residual covariance matrix extracted for usable clip i after fitting to standardized observed channels, and let V_u be the number of usable clip matrices. Standardized inputs do not force residual variances to equal one, so Ψ_i^G is not a correlation matrix. The averaged matrix is

$$\bar{\Psi}_G = \frac{1}{V_u} \sum_{i:U_i=1} \Psi_i^G,$$

taken over usable clips only, and the graph supplied to community detection is

$$\mathbf{W}_G = \left[\operatorname{offdiag} \left\{ \frac{\bar{\Psi}_G + \bar{\Psi}_G^\top}{2} \right\} \right]_+ .$$

This procedure produces a positive residual-covariance graph. It is neither a path graph nor a partial-correlation graph. The resulting graph depends on path search, clip-level model selection, convergence, extraction, averaging, and graph conversion. We therefore treat GIMME as an indirect graph-conversion method without a closed-form block edge w_f^G.

Worst-case accuracy also charges a method for failing to complete. For any method, let U denote the event that a usable fitted graph exists and H the event that the returned partition is correct, conditional on usability. The worst-case success event is

$$H^* = U \cap H, \qquad P(H^*) = P(U)\, P(H \mid U).$$

Failures and timeouts count as non-hits in $P(H^*)$, but they do not enter $P(H \mid U)$. The primary Louvain results make this decomposition concrete. DynEGA returned a usable network for every simulated dataset in the main benchmark, so its worst-case and conditional correct-dimension rates both equal 0.939. GraphicalVAR was usable for 0.941 of datasets; its conditional correct-dimension rate was 0.826 and its worst-case accuracy was 0.777. GIMME was usable for 0.224 of datasets; its conditional correct-dimension rate was 0.783, but its worst-case accuracy was 0.176. Static EGA was usable for 0.997 of datasets and reached a worst-case accuracy of 0.884. A high conditional rate therefore does not imply high worst-case accuracy when a method often fails to return a network at all.

B.6 Method-Comparison Summary

Table B2: Population signal and perturbation sources for the compared recovery methods under the simplified LOMM block target.

Method	Population signal	Main perturbation	Expected implication
DynEGA	$w_f^P(\theta_{\mathrm{GLLA},f})$: partial-correlation mapping of GLLA derivatives	Derivative estimation, embedding selection, glasso regularization, finite N, T	Filtered signal-to-noise depends on the scenario and embedding; any advantage over the lag-1 residual analysis requires evidence from oracle or realized filtered covariances
GraphicalVAR	$w_f^P(\theta_{\mathrm{VAR},f}^{\mathrm{centered}})$: partial-correlation mapping of lag-1 projection residuals	Within-clip centering, VAR estimation, regularization, lag-1 projection attenuation, timeouts	Under the stated symmetric moment assumptions, the centered lag-1 residual covariance preserves the diagonal-plus-rank-one form and follows the shared partial-correlation compression map on its positive branch
GIMME	No closed-form block signal for the implemented graph conversion	Path search, clip-level fitting, extraction, averaging, timeouts	Recovery combines fit usability with a documented method-specific residual-covariance graph conversion

Appendix C: Full Results of the Simulated-versus-Real Comparison

The complete comparison of all eight configurations is reported below for both analyses. The tables retain the bootstrap means, estimated biases, basic reverse-percentile intervals, and percentile intervals so that the displacement can be inspected directly. These quantities diagnose the behavior of the clip-resampling distribution; they are not treated as uncertainty intervals around the absolute plug-in profile discrepancies. The following table then separates the empirical-baseline plus clip-scale profile into its eight component metrics. This second table preserves discrepancies that are obscured by the composite mean, including the remaining differences in amplitude dispersion and first-difference variability.

Table C1: Discrepancy between simulated and real score series for the eight configurations, complete frames analysis. Each estimate averages eight summary statistics within a representation across five equally weighted folds, 100 held-out video clips, and 100 simulated datasets per fold and configuration. Smaller values indicate closer agreement with the real scores. Bootstrap bias is the bootstrap mean minus the estimate. Basic reverse-percentile and percentile intervals are reported as diagnostics rather than as uncertainty intervals for individual configurations. They are conditional on the settings selected within each training fold.

Representation	Configuration	Mean	MCSE	Bootstrap mean	Bias	Basic 95% interval	Percentile 95% interval
Scores	Identical clips	0.3236	0.0007	0.3455	0.0219	[0.2691, 0.3322]	[0.3151, 0.3782]
Scores	Clip baseline	0.2353	0.0009	0.2723	0.0370	[0.1686, 0.2268]	[0.2439, 0.3020]
Scores	Clip scale	0.2877	0.0010	0.3233	0.0356	[0.2197, 0.2803]	[0.2951, 0.3557]
Scores	Clip baseline + scale	0.2218	0.0013	0.2701	0.0484	[0.1433, 0.1992]	[0.2444, 0.3002]
Scores	Activity transition	0.3571	0.0012	0.3830	0.0259	[0.2898, 0.3674]	[0.3468, 0.4245]
Scores	Clip baseline + activity transition	0.2704	0.0013	0.3090	0.0387	[0.1958, 0.2627]	[0.2780, 0.3449]
Scores	Clip scale + activity transition	0.3385	0.0013	0.3708	0.0324	[0.2641, 0.3424]	[0.3345, 0.4128]
Scores	Clip baseline + scale + activity transition	0.2718	0.0014	0.3149	0.0431	[0.1891, 0.2618]	[0.2817, 0.3545]
Log ratio	Identical clips	0.3263	0.0011	0.3538	0.0275	[0.2582, 0.3346]	[0.3180, 0.3944]
Log ratio	Clip baseline	0.3178	0.0011	0.3476	0.0298	[0.2478, 0.3241]	[0.3115, 0.3878]
Log ratio	Clip scale	0.3000	0.0023	0.3453	0.0453	[0.2180, 0.2851]	[0.3149, 0.3821]
Log ratio	Clip baseline + scale	0.2915	0.0023	0.3392	0.0476	[0.2065, 0.2744]	[0.3087, 0.3766]
Log ratio	Activity transition	0.3420	0.0013	0.3778	0.0358	[0.2604, 0.3447]	[0.3393, 0.4236]
Log ratio	Clip baseline + activity transition	0.3331	0.0013	0.3713	0.0382	[0.2490, 0.3327]	[0.3335, 0.4172]
Log ratio	Clip scale + activity transition	0.3224	0.0018	0.3695	0.0470	[0.2288, 0.3134]	[0.3315, 0.4160]
Log ratio	Clip baseline + scale + activity transition	0.3218	0.0019	0.3699	0.0481	[0.2294, 0.3104]	[0.3332, 0.4143]

Table C2: Discrepancy between simulated and real score series for the eight configurations, gap interpolated analysis. Each estimate averages eight summary statistics within a representation across five equally weighted folds, 100 held-out video clips, and 100 simulated datasets per fold and configuration. Smaller values indicate closer agreement with the real scores. Bootstrap bias is the bootstrap mean minus the estimate. Basic reverse-percentile and percentile intervals are reported as diagnostics rather than as uncertainty intervals for individual configurations. They are conditional on the settings selected within each training fold.

Representation	Configuration	Mean	MCSE	Bootstrap mean	Bias	Basic 95% interval	Percentile 95% interval
Scores	Identical clips	0.3164	0.0007	0.3389	0.0226	[0.2614, 0.3231]	[0.3097, 0.3714]
Scores	Clip baseline	0.2290	0.0009	0.2660	0.0369	[0.1618, 0.2183]	[0.2398, 0.2963]
Scores	Clip scale	0.2803	0.0010	0.3168	0.0366	[0.2103, 0.2713]	[0.2892, 0.3502]
Scores	Clip baseline + scale	0.2152	0.0012	0.2639	0.0487	[0.1366, 0.1909]	[0.2395, 0.2939]
Scores	Activity transition	0.3647	0.0012	0.3886	0.0240	[0.2996, 0.3775]	[0.3518, 0.4297]
Scores	Clip baseline + activity transition	0.2797	0.0014	0.3151	0.0353	[0.2075, 0.2763]	[0.2832, 0.3520]
Scores	Clip scale + activity transition	0.3461	0.0014	0.3768	0.0307	[0.2733, 0.3526]	[0.3395, 0.4188]
Scores	Clip baseline + scale + activity transition	0.2819	0.0015	0.3222	0.0403	[0.2016, 0.2752]	[0.2887, 0.3623]
Log ratio	Identical clips	0.3209	0.0011	0.3485	0.0276	[0.2531, 0.3296]	[0.3123, 0.3888]
Log ratio	Clip baseline	0.3124	0.0011	0.3424	0.0300	[0.2433, 0.3188]	[0.3061, 0.3815]
Log ratio	Clip scale	0.2954	0.0023	0.3404	0.0450	[0.2145, 0.2819]	[0.3089, 0.3763]
Log ratio	Clip baseline + scale	0.2869	0.0023	0.3342	0.0474	[0.2036, 0.2710]	[0.3027, 0.3701]
Log ratio	Activity transition	0.3528	0.0015	0.3864	0.0335	[0.2725, 0.3585]	[0.3471, 0.4331]
Log ratio	Clip baseline + activity transition	0.3439	0.0015	0.3799	0.0360	[0.2617, 0.3466]	[0.3413, 0.4261]
Log ratio	Clip scale + activity transition	0.3337	0.0019	0.3782	0.0444	[0.2418, 0.3280]	[0.3395, 0.4257]
Log ratio	Clip baseline + scale + activity transition	0.3360	0.0020	0.3812	0.0452	[0.2432, 0.3288]	[0.3431, 0.4287]

Table C3: Discrepancy for each summary statistic under the baseline-and-scale configuration, complete frames analysis. Results are shown for both representations; smaller values indicate closer agreement with the real scores. For the six single-channel statistics, a unit is one clip-channel summary; for effective rank and total variance, a unit is one clip-level summary. Unit columns report the range across folds and simulation repetitions. The bootstrap means, bias estimates, and 95% intervals are diagnostics rather than uncertainty intervals for the individual discrepancies. They are conditional on the settings selected within each fold.

Representation	Metric	Mean	MCSE	Bias	Basic 95% interval	Percentile 95% interval	Empirical units/fold	Simulated units/fold
Scores	Level median	0.1434	0.0020	0.0379	[0.0657, 0.1364]	[0.1503, 0.2210]	220	220
Scores	Level IQR	0.1456	0.0020	0.0387	[0.0748, 0.1331]	[0.1581, 0.2163]	220	220
Scores	First-difference IQR	0.1846	0.0024	0.0244	[0.1235, 0.1923]	[0.1768, 0.2456]	220	220
Scores	Lag-1 autocorrelation	0.2143	0.0016	0.0241	[0.1221, 0.2419]	[0.1867, 0.3064]	220	220
Scores	Run-length entropy	0.1160	0.0018	0.0396	[0.0467, 0.1029]	[0.1290, 0.1852]	220	220
Scores	Spectral entropy	0.2978	0.0031	0.0139	[0.2122, 0.3513]	[0.2444, 0.3835]	209–220	209–220
Scores	Effective rank	0.3227	0.0033	0.0745	[0.1595, 0.3252]	[0.3202, 0.4859]	20	20
Scores	Total variance	0.3499	0.0058	0.1337	[0.0726, 0.3002]	[0.3995, 0.6271]	20	20
Log ratio	Level median	0.0701	0.0008	0.0250	[0.0322, 0.0559]	[0.0843, 0.1081]	220	220
Log ratio	Level IQR	0.2969	0.0054	0.0497	[0.1777, 0.3025]	[0.2913, 0.4161]	220	220
Log ratio	First-difference IQR	0.4423	0.0073	0.0340	[0.3177, 0.4903]	[0.3943, 0.5669]	220	220
Log ratio	Lag-1 autocorrelation	0.2903	0.0021	0.0206	[0.1778, 0.3465]	[0.2342, 0.4028]	220	220
Log ratio	Run-length entropy	0.1323	0.0020	0.0343	[0.0608, 0.1305]	[0.1340, 0.2037]	220	220
Log ratio	Spectral entropy	0.3979	0.0035	0.0083	[0.3031, 0.4734]	[0.3224, 0.4927]	209–220	209–220
Log ratio	Effective rank	0.3455	0.0024	0.0588	[0.1977, 0.3672]	[0.3237, 0.4932]	20	20
Log ratio	Total variance	0.3570	0.0068	0.1505	[0.1074, 0.2706]	[0.4435, 0.6067]	20	20

Table C4: Discrepancy for each summary statistic under the baseline-and-scale configuration, gap interpolated analysis. Results are shown for both representations; smaller values indicate closer agreement with the real scores. For the six single-channel statistics, a unit is one clip-channel summary; for effective rank and total variance, a unit is one clip-level summary. Unit columns report the range across folds and simulation repetitions. The bootstrap means, bias estimates, and 95% intervals are diagnostics rather than uncertainty intervals for the individual discrepancies. They are conditional on the settings selected within each fold.

Representation	Metric	Mean	MCSE	Bias	Basic 95% interval	Percentile 95% interval	Empirical units/fold	Simulated units/fold
Scores	Level median	0.1431	0.0020	0.0377	[0.0651, 0.1366]	[0.1495, 0.2210]	220	220
Scores	Level IQR	0.1417	0.0020	0.0397	[0.0713, 0.1281]	[0.1554, 0.2122]	220	220
Scores	First-difference IQR	0.1879	0.0025	0.0232	[0.1291, 0.1966]	[0.1792, 0.2467]	220	220
Scores	Lag-1 autocorrelation	0.1937	0.0016	0.0236	[0.1061, 0.2183]	[0.1691, 0.2813]	220	220
Scores	Run-length entropy	0.1179	0.0018	0.0354	[0.0509, 0.1096]	[0.1262, 0.1849]	220	220
Scores	Spectral entropy	0.2724	0.0027	0.0187	[0.1905, 0.3075]	[0.2374, 0.3543]	220	220
Scores	Effective rank	0.3225	0.0033	0.0747	[0.1601, 0.3244]	[0.3205, 0.4848]	20	20
Scores	Total variance	0.3426	0.0058	0.1362	[0.0593, 0.2877]	[0.3975, 0.6259]	20	20
Log ratio	Level median	0.0687	0.0008	0.0258	[0.0301, 0.0536]	[0.0839, 0.1073]	220	220
Log ratio	Level IQR	0.3032	0.0054	0.0462	[0.1867, 0.3120]	[0.2943, 0.4197]	220	220
Log ratio	First-difference IQR	0.4639	0.0075	0.0321	[0.3462, 0.5079]	[0.4199, 0.5816]	220	220
Log ratio	Lag-1 autocorrelation	0.2645	0.0019	0.0234	[0.1505, 0.3161]	[0.2129, 0.3786]	220	220
Log ratio	Run-length entropy	0.1280	0.0021	0.0334	[0.0528, 0.1272]	[0.1288, 0.2032]	220	220
Log ratio	Spectral entropy	0.3674	0.0033	0.0095	[0.2796, 0.4355]	[0.2993, 0.4553]	220	220
Log ratio	Effective rank	0.3452	0.0024	0.0584	[0.1971, 0.3662]	[0.3242, 0.4932]	20	20
Log ratio	Total variance	0.3541	0.0068	0.1502	[0.1100, 0.2658]	[0.4424, 0.5982]	20	20

Appendix D: Grouping and Trajectory Diagnostics Among Usable Fits

Table D1: Louvain grouping diagnostics for the main benchmark with $T = 100$. Each estimate uses the datasets available for it. Monte Carlo standard errors are calculated per simulated dataset, and intervals resample whole datasets, stratified by condition.

Method	Diagnostic	Estimate	MCSE	95% interval	Datasets	Usable fits	Correct-dimensional fits
DynEGA	Signed dimension error	0.166	0.0057	[0.154, 0.177]	21600	21600	20282
DynEGA	Absolute dimension error	0.168	0.0057	[0.157, 0.180]	21600	21600	20282
DynEGA	Adjusted Rand index	0.935	0.0015	[0.932, 0.938]	21600	21600	20282
DynEGA	Variation of information	0.095	0.0023	[0.091, 0.100]	21600	21600	20282
DynEGA	Assignment accuracy given usable	0.965	0.0009	[0.963, 0.966]	21600	21600	20282
GraphicalVAR	Signed dimension error	0.979	0.0171	[0.947, 1.011]	20329	20329	16784
GraphicalVAR	Absolute dimension error	0.979	0.0171	[0.945, 1.012]	20329	20329	16784
GraphicalVAR	Adjusted Rand index	0.867	0.0022	[0.863, 0.872]	20329	20329	16784
GraphicalVAR	Variation of information	0.183	0.0031	[0.177, 0.189]	20329	20329	16784
GraphicalVAR	Assignment accuracy given usable	0.902	0.0016	[0.899, 0.906]	20329	20329	16784
GIMME	Signed dimension error	0.319	0.0110	[0.298, 0.341]	4846	4846	3795
GIMME	Absolute dimension error	0.319	0.0110	[0.298, 0.341]	4846	4846	3795
GIMME	Adjusted Rand index	0.733	0.0057	[0.722, 0.744]	4846	4846	3795
GIMME	Variation of information	0.337	0.0073	[0.323, 0.351]	4846	4846	3795
GIMME	Assignment accuracy given usable	0.877	0.0027	[0.872, 0.882]	4846	4846	3795
Static EGA	Signed dimension error	0.799	0.0159	[0.770, 0.829]	21526	21526	19093
Static EGA	Absolute dimension error	0.799	0.0159	[0.767, 0.829]	21526	21526	19093
Static EGA	Adjusted Rand index	0.894	0.0021	[0.889, 0.898]	21526	21526	19093
Static EGA	Variation of information	0.147	0.0029	[0.142, 0.153]	21526	21526	19093
Static EGA	Assignment accuracy given usable	0.922	0.0015	[0.919, 0.925]	21526	21526	19093
Block oracle	Signed dimension error	0.000	0.0000	[0.000, 0.000]	21600	21600	21600
Block oracle	Absolute dimension error	0.000	0.0000	[0.000, 0.000]	21600	21600	21600
Block oracle	Adjusted Rand index	1.000	0.0000	[1.000, 1.000]	21600	21600	21600
Block oracle	Variation of information	0.000	0.0000	[0.000, 0.000]	21600	21600	21600
Block oracle	Assignment accuracy given usable	1.000	0.0000	[1.000, 1.000]	21600	21600	21600

Table D2: Louvain grouping diagnostics for the main benchmark with $T = 25$. Each estimate uses the datasets available for it. Monte Carlo standard errors are calculated per simulated dataset, and intervals resample whole datasets, stratified by condition.

Method	Diagnostic	Estimate	MCSE	95% interval	Datasets	Usable fits	Correct-dimensional fits
DynEGA	Signed dimension error	0.492	0.0102	[0.471, 0.512]	21600	21600	18636
DynEGA	Absolute dimension error	0.494	0.0102	[0.474, 0.515]	21600	21600	18636
DynEGA	Adjusted Rand index	0.902	0.0018	[0.899, 0.906]	21600	21600	18636
DynEGA	Variation of information	0.137	0.0025	[0.132, 0.143]	21600	21600	18636
DynEGA	Assignment accuracy given usable	0.935	0.0012	[0.933, 0.938]	21600	21600	18636
GraphicalVAR	Signed dimension error	1.048	0.0154	[1.017, 1.076]	21579	21579	16641
GraphicalVAR	Absolute dimension error	1.048	0.0154	[1.016, 1.080]	21579	21579	16641
GraphicalVAR	Adjusted Rand index	0.859	0.0020	[0.855, 0.863]	21579	21579	16641
GraphicalVAR	Variation of information	0.200	0.0028	[0.195, 0.206]	21579	21579	16641
GraphicalVAR	Assignment accuracy given usable	0.893	0.0015	[0.890, 0.896]	21579	21579	16641
Static EGA	Signed dimension error	0.576	0.0136	[0.551, 0.603]	21541	21541	19721
Static EGA	Absolute dimension error	0.576	0.0136	[0.550, 0.603]	21541	21541	19721
Static EGA	Adjusted Rand index	0.922	0.0018	[0.919, 0.926]	21541	21541	19721
Static EGA	Variation of information	0.108	0.0025	[0.103, 0.113]	21541	21541	19721
Static EGA	Assignment accuracy given usable	0.943	0.0013	[0.940, 0.945]	21541	21541	19721
Block oracle	Signed dimension error	0.000	0.0000	[0.000, 0.000]	21600	21600	21600
Block oracle	Absolute dimension error	0.000	0.0000	[0.000, 0.000]	21600	21600	21600
Block oracle	Adjusted Rand index	1.000	0.0000	[1.000, 1.000]	21600	21600	21600
Block oracle	Variation of information	0.000	0.0000	[0.000, 0.000]	21600	21600	21600
Block oracle	Assignment accuracy given usable	1.000	0.0000	[1.000, 1.000]	21600	21600	21600

Table D3: Louvain trajectory diagnostics for the main benchmark with $T = 100$. Correct-dimension rows summarize usable fits that found the right number of dimensions. Marginal rows assign correlation 0 and NRMSE 2 to unusable fits. Higher correlation and lower NRMSE indicate closer recovery of the activity-scaled latent trajectories.

Method	Diagnostic	Estimate	MCSE	95% interval	Datasets	Usable fits	Correct-dimensional fits
DynEGA	Score correlation (status-aware marginal)	0.549	0.0014	[0.546, 0.552]	21600	21600	20282
DynEGA	Score NRMSE (status-aware marginal)	1.435	0.0013	[1.432, 1.438]	21600	21600	20282
GraphicalVAR	Score correlation (status-aware marginal)	0.668	0.0020	[0.664, 0.671]	21600	20329	16784
GraphicalVAR	Score NRMSE (status-aware marginal)	0.753	0.0036	[0.746, 0.760]	21600	20329	16784
GIMME	Score correlation (status-aware marginal)	0.132	0.0019	[0.129, 0.136]	21600	4846	3795
GIMME	Score NRMSE (status-aware marginal)	1.749	0.0034	[1.742, 1.756]	21600	4846	3795
Static EGA	Score correlation (status-aware marginal)	0.727	0.0017	[0.723, 0.730]	21600	21526	19093
Static EGA	Score NRMSE (status-aware marginal)	0.641	0.0031	[0.635, 0.647]	21600	21526	19093
Block oracle	Score correlation (status-aware marginal)	0.729	0.0017	[0.726, 0.733]	21600	21600	21600
Block oracle	Score NRMSE (status-aware marginal)	0.642	0.0031	[0.636, 0.649]	21600	21600	21600
DynEGA	Score correlation (correct-dimensional usable fits)	0.574	0.0013	[0.572, 0.577]	20282	21600	20282
DynEGA	Score NRMSE (correct-dimensional usable fits)	1.418	0.0014	[1.416, 1.421]	20282	21600	20282
GraphicalVAR	Score correlation (correct-dimensional usable fits)	0.791	0.0014	[0.789, 0.794]	16784	20329	16784
GraphicalVAR	Score NRMSE (correct-dimensional usable fits)	0.568	0.0031	[0.562, 0.574]	16784	20329	16784
GIMME	Score correlation (correct-dimensional usable fits)	0.606	0.0034	[0.600, 0.613]	3795	4846	3795
GIMME	Score NRMSE (correct-dimensional usable fits)	0.898	0.0050	[0.888, 0.907]	3795	4846	3795
Static EGA	Score correlation (correct-dimensional usable fits)	0.787	0.0014	[0.784, 0.789]	19093	21526	19093
Static EGA	Score NRMSE (correct-dimensional usable fits)	0.563	0.0031	[0.557, 0.569]	19093	21526	19093
Block oracle	Score correlation (correct-dimensional usable fits)	0.729	0.0017	[0.726, 0.732]	21600	21600	21600
Block oracle	Score NRMSE (correct-dimensional usable fits)	0.642	0.0031	[0.636, 0.648]	21600	21600	21600

Table D4: Louvain trajectory diagnostics for the main benchmark with $T = 25$. Correct-dimension rows summarize usable fits that found the right number of dimensions. Marginal rows assign correlation 0 and NRMSE 2 to unusable fits. Higher correlation and lower NRMSE indicate closer recovery of the activity-scaled latent trajectories.

Method	Diagnostic	Estimate	MCSE	95% interval	Datasets	Usable fits	Correct-dimensional fits
DynEGA	Score correlation (status-aware marginal)	0.611	0.0011	[0.608, 0.613]	21600	21600	18636
DynEGA	Score NRMSE (status-aware marginal)	1.412	0.0012	[1.409, 1.414]	21600	21600	18636
GraphicalVAR	Score correlation (status-aware marginal)	0.799	0.0012	[0.796, 0.801]	21600	21579	16641
GraphicalVAR	Score NRMSE (status-aware marginal)	0.587	0.0023	[0.582, 0.591]	21600	21579	16641
GIMME	Score correlation (status-aware marginal)	0.000	0.0000	[0.000, 0.000]	21600	0	0
GIMME	Score NRMSE (status-aware marginal)	2.000	0.0000	[2.000, 2.000]	21600	0	0
Static EGA	Score correlation (status-aware marginal)	0.801	0.0012	[0.799, 0.803]	21600	21541	19721
Static EGA	Score NRMSE (status-aware marginal)	0.595	0.0024	[0.591, 0.600]	21600	21541	19721
Block oracle	Score correlation (status-aware marginal)	0.798	0.0012	[0.796, 0.801]	21600	21600	21600
Block oracle	Score NRMSE (status-aware marginal)	0.601	0.0024	[0.596, 0.605]	21600	21600	21600
DynEGA	Score correlation (correct-dimensional usable fits)	0.654	0.0010	[0.652, 0.655]	18636	21600	18636
DynEGA	Score NRMSE (correct-dimensional usable fits)	1.382	0.0013	[1.380, 1.385]	18636	21600	18636
GraphicalVAR	Score correlation (correct-dimensional usable fits)	0.871	0.0008	[0.870, 0.873]	16641	21579	16641
GraphicalVAR	Score NRMSE (correct-dimensional usable fits)	0.466	0.0022	[0.461, 0.470]	16641	21579	16641
Static EGA	Score correlation (correct-dimensional usable fits)	0.836	0.0009	[0.834, 0.838]	19721	21541	19721
Static EGA	Score NRMSE (correct-dimensional usable fits)	0.548	0.0023	[0.544, 0.553]	19721	21541	19721
Block oracle	Score correlation (correct-dimensional usable fits)	0.798	0.0012	[0.796, 0.801]	21600	21600	21600
Block oracle	Score NRMSE (correct-dimensional usable fits)	0.601	0.0024	[0.596, 0.605]	21600	21600	21600

Appendix E: Runtime and Feasibility Accounting

Table E1: Runtime and feasibility per fit in the main benchmark. Completion intervals resample whole simulated datasets. Runtime summaries use successful fits and include estimating the structure plus extracting the network and partitioning. Each successful structure is reused for the three community-algorithm analyses.

Analysis	Time-series length	Method	Structure fits	Successful fits	Timeout fits	Error fits	Completion	Runtime median [IQR], s
Matched benchmark	$T = 100$	DynEGA	21600	21600	0	0	1.000 [1.000, 1.000]	3.69 [3.18, 4.25]
Matched benchmark	$T = 100$	GIMME	21600	4846	14477	2277	0.224 [0.219, 0.230]	518.16 [398.98, 603.71]
Matched benchmark	$T = 100$	GraphicalVAR	21600	20329	1271	0	0.941 [0.938, 0.944]	85.41 [53.82, 185.91]
Matched benchmark	$T = 100$	Block oracle	21600	21600	0	0	1.000 [1.000, 1.000]	0.06 [0.03, 0.08]
Matched benchmark	$T = 100$	Static EGA	21600	21526	0	74	0.997 [0.996, 0.997]	2.15 [2.05, 2.26]
Matched benchmark	$T = 25$	DynEGA	21600	21600	0	0	1.000 [1.000, 1.000]	2.28 [2.13, 2.52]
Matched benchmark	$T = 25$	GIMME	21600	0	0	21600	0.000 [0.000, 0.000]	Not available
Matched benchmark	$T = 25$	GraphicalVAR	21600	21579	21	0	0.999 [0.999, 0.999]	66.49 [33.96, 169.39]
Matched benchmark	$T = 25$	Block oracle	21600	21600	0	0	1.000 [1.000, 1.000]	0.04 [0.03, 0.05]
Matched benchmark	$T = 25$	Static EGA	21600	21541	0	59	0.997 [0.997, 0.998]	2.20 [2.10, 2.32]

Appendix F: Scenario Parameter Values

The scenario families use the numeric settings below. The displayed σ_q vectors are the per-step shock standard deviations used in the analyses. The 100-observation benchmark uses $\Delta = 0.4$. Conditions with 300 observations occur only in the broad DynEGA extension and use $\Delta = 0.25$, so these conditions differ in the number of observations, the interval between them, the total time spanned, and discrete shock specification.

Codes A–D use asymptotically stable feedback. Codes E–F use $\eta_f < 0$ and $\zeta_f > 0$, which produces oscillatory finite-horizon amplification. Their interpretation is restricted to the stated observation horizons. The empirical compositional calibration uses stable negative-ζ dynamics and is analyzed separately. All oscillatory factors satisfy the cadence restriction $0 < \omega_f < \pi/\Delta$ at the observation intervals used here.

Table F1: Generator-only amplitude diagnostics for the finite-horizon amplification families with $N = 50$, four indicators per dimension, loading 0.60, residual standard deviation 0.25, and 25 replications. The final-tenth energy fraction is the proportion of squared activity-scaled latent signal occurring in the last 10% of observations, averaged across clips within a panel and then summarized across replications.

Scenario	Dimensions	T	H	Median max $\lvert\tilde{x}\rvert$	95th percentile max $\lvert\tilde{x}\rvert$	Median observed IQR	Final-tenth energy fraction
Ranked finite-horizon amplification	2	100	39.60	55.9	61.5	3.47	0.316
Ranked finite-horizon amplification	2	300	74.75	329.4	376.9	7.46	0.338
Ranked finite-horizon amplification	3	100	39.60	62.6	73.0	1.90	0.368
Ranked finite-horizon amplification	3	300	74.75	430.3	478.6	4.58	0.428
Ranked finite-horizon amplification	4	100	39.60	85.6	102.7	1.57	0.404
Ranked finite-horizon amplification	4	300	74.75	994.9	1092.7	4.02	0.498
Switching finite-horizon amplification	2	100	39.60	69.5	81.9	4.71	0.319
Switching finite-horizon amplification	2	300	74.75	388.1	491.3	9.31	0.324
Switching finite-horizon amplification	3	100	39.60	77.7	87.5	2.33	0.367
Switching finite-horizon amplification	3	300	74.75	518.5	608.5	5.74	0.448
Switching finite-horizon amplification	4	100	39.60	105.2	122.2	1.92	0.402
Switching finite-horizon amplification	4	300	74.75	1201.6	1350.2	4.97	0.517

Table F2: Activity schedules and dynamic-feedback settings for each scenario family.

Code	Family	Dynamic family	Switching	Gain	Dwell	Velocity feedback
A	Low-amplitude single dominance	Stable template	No	0.35 / 0.05	1.00 / 1.00	Negative
B	High-amplitude single dominance	Stable template	No	1.00 / 0.08	1.00 / 1.00	Negative
C	High-amplitude switching	Stable template	Yes	1.00 / 0.08	0.20 / 0.45	Negative
D	High-amplitude rapid switching	Stable template	Yes	1.25 / 0.08	0.03 / 0.10	Negative
E	Ranked finite-horizon amplification	Amplification	No	1.00 / 0.08	1.00 / 1.00	Positive
F	Switching finite-horizon amplification	Amplification	Yes	1.10 / 0.10	0.12 / 0.30	Positive

Table F3: Dynamic parameters for each scenario family. Acceleration-input SD is the per-step standard deviation σ_q used in each condition.

Code	Dimensions	eta	zeta	Acceleration-input SD
A	2D	-0.30; -0.55	-1.70; -1.00	0.06; 0.15
A	3D	-0.30; -0.55; -0.85	-1.70; -1.00; -0.65	0.06; 0.15; 0.23
A	4D	-0.30; -0.55; -0.85; -1.10	-1.70; -1.00; -0.65; -0.30	0.06; 0.15; 0.23; 0.33
B	2D	-0.30; -0.55	-1.70; -1.00	0.12; 0.30
B	3D	-0.30; -0.55; -0.85	-1.70; -1.00; -0.65	0.12; 0.30; 0.45
B	4D	-0.30; -0.55; -0.85; -1.10	-1.70; -1.00; -0.65; -0.30	0.12; 0.30; 0.45; 0.65
C	2D	-0.30; -0.55	-1.70; -1.00	0.12; 0.30
C	3D	-0.30; -0.55; -0.85	-1.70; -1.00; -0.65	0.12; 0.30; 0.45
C	4D	-0.30; -0.55; -0.85; -1.10	-1.70; -1.00; -0.65; -0.30	0.12; 0.30; 0.45; 0.65
D	2D	-0.30; -0.55	-1.70; -1.00	0.15; 0.38
D	3D	-0.30; -0.55; -0.85	-1.70; -1.00; -0.65	0.15; 0.38; 0.56
D	4D	-0.30; -0.55; -0.85; -1.10	-1.70; -1.00; -0.65; -0.30	0.15; 0.38; 0.56; 0.81
E	2D	-0.05; -0.10	0.08; 0.10	0.25; 0.25
E	3D	-0.05; -0.10; -0.15	0.08; 0.10; 0.12	0.25; 0.25; 0.25
E	4D	-0.05; -0.10; -0.15; -0.20	0.08; 0.10; 0.12; 0.14	0.25; 0.25; 0.25; 0.25
F	2D	-0.05; -0.10	0.08; 0.10	0.35; 0.35
F	3D	-0.05; -0.10; -0.15	0.08; 0.10; 0.12	0.35; 0.35; 0.35
F	4D	-0.05; -0.10; -0.15; -0.20	0.08; 0.10; 0.12; 0.14	0.35; 0.35; 0.35; 0.35

Table F4: Deterministic stability diagnostics and modal exponential-rate factors $g(H) = \exp\{H \max \mathrm{Re}(r)\}$ for each scenario factor. The $g(H)$ columns describe modal rates over the observed horizon rather than total finite-horizon state amplification. $H[100] = 39.6$ and $H[300] = 74.75$ under the stated record designs.

Code	Dimensions	Factor	eta	zeta	Discriminant	Max Re(root)	rho(Phi[.4])	rho(Phi[.25])	g(H[100])	g(H[300])	Regime
A	2D	1	-0.30	-1.70	1.6900	-0.200	0.9231	0.9512	0.0004	0.0000	Stable nonoscillatory
A	2D	2	-0.55	-1.00	-1.2000	-0.500	0.8187	0.8825	0.0000	0.0000	Stable oscillatory
A	3D	1	-0.30	-1.70	1.6900	-0.200	0.9231	0.9512	0.0004	0.0000	Stable nonoscillatory
A	3D	2	-0.55	-1.00	-1.2000	-0.500	0.8187	0.8825	0.0000	0.0000	Stable oscillatory
A	3D	3	-0.85	-0.65	-2.9775	-0.325	0.8781	0.9220	0.0000	0.0000	Stable oscillatory
A	4D	1	-0.30	-1.70	1.6900	-0.200	0.9231	0.9512	0.0004	0.0000	Stable nonoscillatory
A	4D	2	-0.55	-1.00	-1.2000	-0.500	0.8187	0.8825	0.0000	0.0000	Stable oscillatory
A	4D	3	-0.85	-0.65	-2.9775	-0.325	0.8781	0.9220	0.0000	0.0000	Stable oscillatory
A	4D	4	-1.10	-0.30	-4.3100	-0.150	0.9418	0.9632	0.0026	0.0000	Stable oscillatory
B	2D	1	-0.30	-1.70	1.6900	-0.200	0.9231	0.9512	0.0004	0.0000	Stable nonoscillatory
B	2D	2	-0.55	-1.00	-1.2000	-0.500	0.8187	0.8825	0.0000	0.0000	Stable oscillatory
B	3D	1	-0.30	-1.70	1.6900	-0.200	0.9231	0.9512	0.0004	0.0000	Stable nonoscillatory
B	3D	2	-0.55	-1.00	-1.2000	-0.500	0.8187	0.8825	0.0000	0.0000	Stable oscillatory
B	3D	3	-0.85	-0.65	-2.9775	-0.325	0.8781	0.9220	0.0000	0.0000	Stable oscillatory
B	4D	1	-0.30	-1.70	1.6900	-0.200	0.9231	0.9512	0.0004	0.0000	Stable nonoscillatory
B	4D	2	-0.55	-1.00	-1.2000	-0.500	0.8187	0.8825	0.0000	0.0000	Stable oscillatory
B	4D	3	-0.85	-0.65	-2.9775	-0.325	0.8781	0.9220	0.0000	0.0000	Stable oscillatory
B	4D	4	-1.10	-0.30	-4.3100	-0.150	0.9418	0.9632	0.0026	0.0000	Stable oscillatory
C	2D	1	-0.30	-1.70	1.6900	-0.200	0.9231	0.9512	0.0004	0.0000	Stable nonoscillatory
C	2D	2	-0.55	-1.00	-1.2000	-0.500	0.8187	0.8825	0.0000	0.0000	Stable oscillatory
C	3D	1	-0.30	-1.70	1.6900	-0.200	0.9231	0.9512	0.0004	0.0000	Stable nonoscillatory
C	3D	2	-0.55	-1.00	-1.2000	-0.500	0.8187	0.8825	0.0000	0.0000	Stable oscillatory
C	3D	3	-0.85	-0.65	-2.9775	-0.325	0.8781	0.9220	0.0000	0.0000	Stable oscillatory

C	4D	1	-0.30	-1.70	1.6900	-0.200	0.9231	0.9512	0.0004	0.0000	Stable nonoscillatory
C	4D	2	-0.55	-1.00	-1.2000	-0.500	0.8187	0.8825	0.0000	0.0000	Stable oscillatory
C	4D	3	-0.85	-0.65	-2.9775	-0.325	0.8781	0.9220	0.0000	0.0000	Stable oscillatory
C	4D	4	-1.10	-0.30	-4.3100	-0.150	0.9418	0.9632	0.0026	0.0000	Stable oscillatory
D	2D	1	-0.30	-1.70	1.6900	-0.200	0.9231	0.9512	0.0004	0.0000	Stable nonoscillatory
D	2D	2	-0.55	-1.00	-1.2000	-0.500	0.8187	0.8825	0.0000	0.0000	Stable oscillatory
D	3D	1	-0.30	-1.70	1.6900	-0.200	0.9231	0.9512	0.0004	0.0000	Stable nonoscillatory
D	3D	2	-0.55	-1.00	-1.2000	-0.500	0.8187	0.8825	0.0000	0.0000	Stable oscillatory
D	3D	3	-0.85	-0.65	-2.9775	-0.325	0.8781	0.9220	0.0000	0.0000	Stable oscillatory
D	4D	1	-0.30	-1.70	1.6900	-0.200	0.9231	0.9512	0.0004	0.0000	Stable nonoscillatory
D	4D	2	-0.55	-1.00	-1.2000	-0.500	0.8187	0.8825	0.0000	0.0000	Stable oscillatory
D	4D	3	-0.85	-0.65	-2.9775	-0.325	0.8781	0.9220	0.0000	0.0000	Stable oscillatory
D	4D	4	-1.10	-0.30	-4.3100	-0.150	0.9418	0.9632	0.0026	0.0000	Stable oscillatory
E	2D	1	-0.05	0.08	-0.1936	0.040	1.0161	1.0101	4.8744	19.8857	Finite-horizon amplifying oscillatory
E	2D	2	-0.10	0.10	-0.3900	0.050	1.0202	1.0126	7.2427	41.9929	Finite-horizon amplifying oscillatory
E	3D	1	-0.05	0.08	-0.1936	0.040	1.0161	1.0101	4.8744	19.8857	Finite-horizon amplifying oscillatory
E	3D	2	-0.10	0.10	-0.3900	0.050	1.0202	1.0126	7.2427	41.9929	Finite-horizon amplifying oscillatory
E	3D	3	-0.15	0.12	-0.5856	0.060	1.0243	1.0151	10.7618	88.6770	Finite-horizon amplifying oscillatory
E	4D	1	-0.05	0.08	-0.1936	0.040	1.0161	1.0101	4.8744	19.8857	Finite-horizon amplifying oscillatory
E	4D	2	-0.10	0.10	-0.3900	0.050	1.0202	1.0126	7.2427	41.9929	Finite-horizon amplifying oscillatory
E	4D	3	-0.15	0.12	-0.5856	0.060	1.0243	1.0151	10.7618	88.6770	Finite-horizon amplifying oscillatory
E	4D	4	-0.20	0.14	-0.7804	0.070	1.0284	1.0177	15.9906	187.2604	Finite-horizon amplifying oscillatory
F	2D	1	-0.05	0.08	-0.1936	0.040	1.0161	1.0101	4.8744	19.8857	Finite-horizon amplifying oscillatory
F	2D	2	-0.10	0.10	-0.3900	0.050	1.0202	1.0126	7.2427	41.9929	Finite-horizon amplifying oscillatory
F	3D	1	-0.05	0.08	-0.1936	0.040	1.0161	1.0101	4.8744	19.8857	Finite-horizon amplifying oscillatory
F	3D	2	-0.10	0.10	-0.3900	0.050	1.0202	1.0126	7.2427	41.9929	Finite-horizon amplifying oscillatory
F	3D	3	-0.15	0.12	-0.5856	0.060	1.0243	1.0151	10.7618	88.6770	Finite-horizon amplifying oscillatory
F	4D	1	-0.05	0.08	-0.1936	0.040	1.0161	1.0101	4.8744	19.8857	Finite-horizon amplifying oscillatory
F	4D	2	-0.10	0.10	-0.3900	0.050	1.0202	1.0126	7.2427	41.9929	Finite-horizon amplifying oscillatory
F	4D	3	-0.15	0.12	-0.5856	0.060	1.0243	1.0151	10.7618	88.6770	Finite-horizon amplifying oscillatory
F	4D	4	-0.20	0.14	-0.7804	0.070	1.0284	1.0177	15.9906	187.2604	Finite-horizon amplifying oscillatory

Appendix G: Community-Detection Sensitivity

Louvain, Leiden, and Walktrap were applied to each retained fitted graph. Their results were closely aligned in the matched designs (Figure G1) and led to the same interpretation of the targeted departures (Figure G2). These comparisons reuse fitted structures and therefore assess the community-detection step rather than estimator variability.

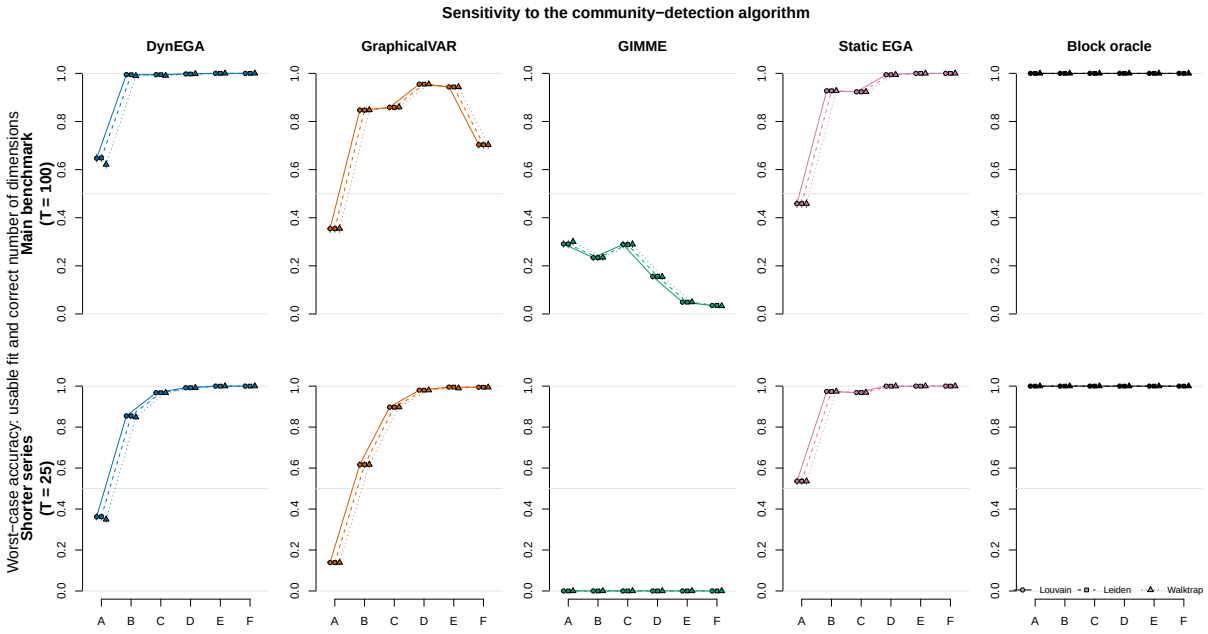


Figure G1: Worst-case accuracy under Louvain, Leiden, and Walktrap in the main benchmark. Each fitted network is reused across all three community-detection algorithms.

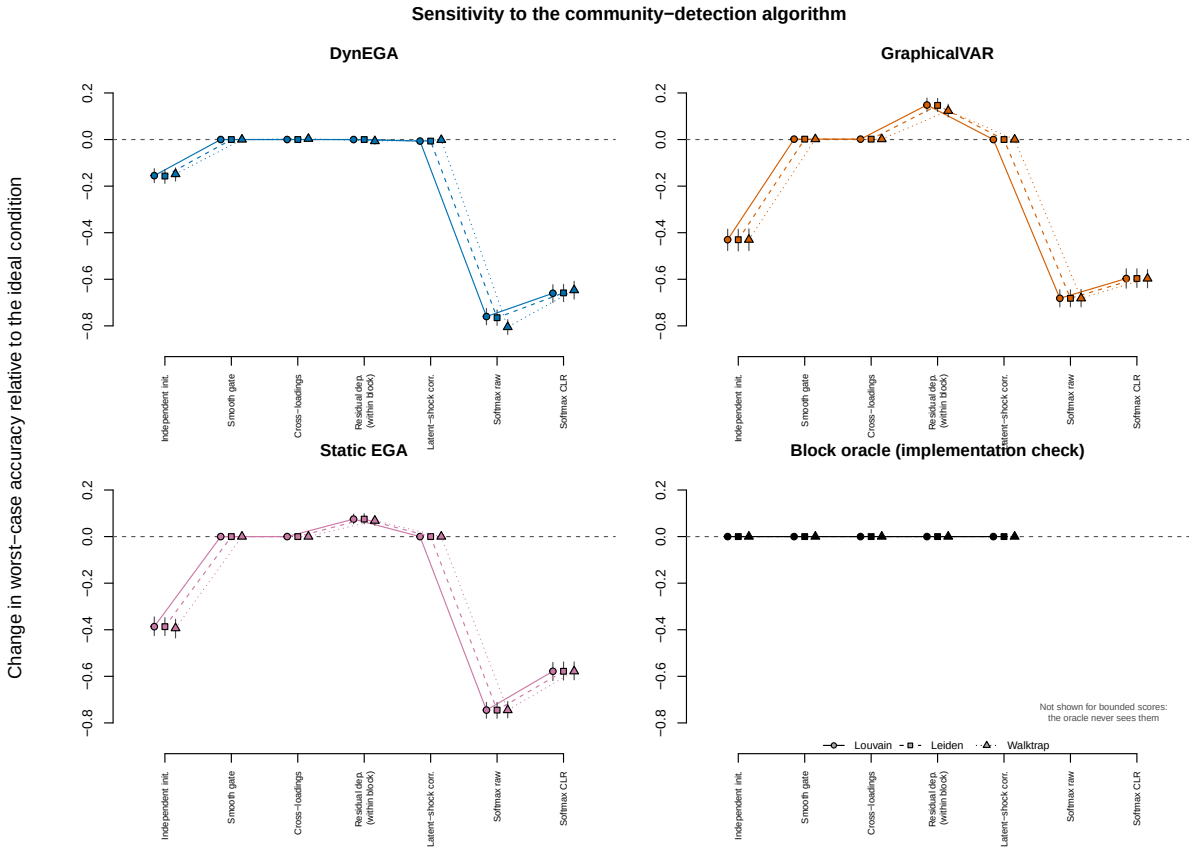


Figure G2: Sensitivity of the paired departure effects to Louvain, Leiden, and Walktrap. Values are changes in worst-case accuracy relative to the ideal condition. The oracle is omitted from the bounded-score and log-ratio comparisons because it receives the true structure directly and therefore says nothing about what happens once scores are constrained to sum to one.